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Special Collection:

Key Hydrological Processes and Its Controlling Factors in Terrestrial Ecosystems

Key Points:

- Machine learning methods can be effectively integrated with hydrological models, resulting in superior streamflow simulation capabilities
- Different controlling factors exerted nonlinear influences on streamflow, with notable differences and threshold effects
- Precipitation does not dominate the interaction effects with land use, with land use change as the primary driver of streamflow variation

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Exploring the Controlling Factors of Watershed Streamflow Variability Using Hydrological and Machine Learning Models

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Abstract Studying streamflow processes and controlling factors is crucial for sustainable water resource management. This study demonstrated the potential of integrating hydrological models with machine learning by constructing two machine learning methods, Extreme Gradient Boosting (XGBoost) and Random Forest (RF), based on the input and output data from the Soil and Water Assessment Tool (SWAT) and comparing their streamflow simulation performances. The Shapley Additive exPlanations (SHAP) method identified the controlling factors and their interactions in streamflow variation, whereas scenario simulations quantified the relative contributions of climate and land use changes. The results showed that when integrated with the SWAT model, XGBoost demonstrated better streamflow simulation performance than RF. Among the key factors influencing streamflow variation, area was the most important, with precipitation having a stronger impact than temperature, positively affecting streamflow when exceeding 550 mm. Different land use types exerted nonlinear impacts on streamflow, with notable differences and threshold effects. Specifically, grassland, cropland, and forest positively contributed to streamflow when their proportions were below 50%, above 20%, and between 30% and 50%, respectively. Nonlinear interaction effects on streamflow between land use types resulted in positive or negative contributions at specific proportion thresholds. Furthermore, precipitation was not dominant in the interaction with land use. Streamflow changes were primarily driven by drastic land use changes, which contributed 55.71%, while climate change accounted for 44.27%. This integration of hydrological models with machine learning revealed the complex impacts of climate and land use changes on streamflow, offering scientific insights for watershed water resource management.

Plain Language Summary As global water resources have increased in scarcity, it has become necessary to study the processes underlying streamflow variation and its controlling factors. In this study, it was revealed that combining hydrological models with machine learning methods is a promising approach for addressing this issue. When integrated with the SWAT model, XGBoost demonstrated better streamflow simulation performance than RF. Through the SHAP method, we identified the key factors influencing streamflow variation. Watershed area was the main factor, and precipitation exerted a greater impact on streamflow than temperature. Different land use types had nonlinear impacts on streamflow with significant threshold effects. The interaction effects between land use types revealed that different land use combinations played complex roles in regulating streamflow. Precipitation did not dominate the interaction effects with land use. Instead, drastic land use changes led to abrupt variations in watershed streamflow, although climate changes played contributing roles as well.

1. Introduction

Streamflow is a critical component of the hydrological cycle (Ok & Kanae, 2006). As global water resources have become increasingly strained, changes in streamflow not only profoundly influence natural ecosystems but also directly influence the sustainable development of human society. Variations in streamflow significantly affect soil and water conservation, agricultural irrigation, flood management, and the supply of drinking water (Ben-Asher & Berliner, 1994; Lane, 2017; Lye, 2009; Ran et al., 2012). By altering precipitation patterns and temperature, climate change significantly impacts watershed streamflow dynamics and distribution (Berghuijs et al., 2017; Ding et al., 2024). Additionally, land use changes within watersheds—particularly the agricultural land expansion, increased urbanization, and changes in vegetation cover—are increasingly influencing streamflow patterns

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(Anderson et al., 2022; Li et al., 2018; Zhang et al., 2018). In many regions, shifts in precipitation patterns, temperature fluctuations, and land use changes have led to instability in watershed streamflow processes, rendering predictions more challenging. Therefore, research on streamflow variations and their driving factors, especially the interactions between climate change and land use, has become pivotal for hydrological studies.

In recent years, numerous researchers have investigated the influences of driving factors on streamflow using various methodologies (Dey & Mishra, 2017). These methodologies primarily include empirical statistical methods, water–heat coupling balance approaches and hydrological models (Wang et al., 2022). Empirical statistical methods, such as regression analysis and the rainfall–streamflow double cumulative curve method, aim to establish statistical relationships between streamflow and relevant influencing variables, which have been used to elucidate the sensitivity levels of streamflow to various climate and human activities (Cheng & Yu, 2019). However, these methods typically rely on extensive observational data and lack in-depth analysis of the physical mechanisms underlying watershed hydrological processes (Xu et al., 2014). Water–heat coupling balance approaches, which are based on the Budyko hypothesis, can be employed to estimate the sensitivity levels of streamflow to climate change and land use change (Jiang et al., 2015; Yang et al., 2014). Although the Budyko method is straightforward to implement, neglecting watershed water storage variations when the Budyko method is applied at annual and subannual scales can lead to significant errors (Gan et al., 2021; Zeng & Cai, 2016). Additionally, the Budyko method fails to account for vegetation coevolution under nonsteady-state conditions (Gan et al., 2021). Traditional hydrological models, such as the Soil and Water Assessment Tool (SWAT), incorporate physical mechanisms, thereby comprehensively accounting for various factors, including meteorology, soil, topography, land use, and human activities, to simulate complex hydrological processes with high accuracy and reliability (Arnold et al., 2012; Gassman et al., 2014). The SWAT model can be used to examine streamflow changes by altering input variables (Luo et al., 2019; Sharma et al., 2022). However, owing to its numerous parameters and high computational demands, model calibration and validation are typically cumbersome and time-consuming processes (Rajib et al., 2016). This high complexity may limit model application in large-scale watersheds or situations requiring rapid decision-making. In summary, although empirical statistical methods, water–heat coupling balance approaches, and traditional hydrological models have made notable progress in streamflow research, these methods still have limitations in terms of their accuracy, mechanistic explanations, and computational efficiency. Consequently, there is an urgent need to develop a more efficient and explanatory simulation method capable of precisely capturing the influences of driving factors on streamflow.

With the rapid advancement in machine learning technologies, methods such as artificial neural networks and ensemble learning have demonstrated robust data-driven capabilities and flexibility in streamflow simulations (Boucher et al., 2020; Zounemat-Kermani et al., 2021). These approaches leverage large volumes of historical observational data to automatically identify and model the complex nonlinear relationships inherent in streamflow processes, thus significantly enhancing the prediction accuracy. Compared with traditional hydrological models, long short-term memory (LSTM) networks have been successfully applied to predict streamflow in multiple watersheds, demonstrating superior performance (Kratzert et al., 2019; Tang et al., 2023). Ensemble learning methods, such as the random forest (RF) and extreme gradient boosting (XGBoost) methods, have also been widely employed in streamflow simulations, and they provide exceptional capabilities in resolving high-dimensional data and capturing complex interactions among features (Li et al., 2020; Wang et al., 2022). These machine learning approaches provide novel perspectives and tools for hydrological research. Nonetheless, traditional machine learning methods are often perceived as black boxes owing to their limited interpretability, thus limiting their application in mechanistic studies of hydrological processes (Nearing et al., 2021). With the rapid advancement in explainable artificial intelligence technologies, the introduction of the SHapley Additive exPlanations (SHAP) method has increased the transparency and interpretability of machine learning models. Grounded in game theory, the SHAP method aims to quantify the contribution of each feature to the prediction outcome by calculating the respective Shapley values, thereby enhancing model interpretability (Das & Rad, 2020; Lundberg, 2017). Through SHAP analysis, researchers can not only enhance model interpretability but also better understand the specific roles and interactions of various driving factors in determining streamflow variations. However, the application of explainable machine learning methods for the attribution analysis of streamflow variables remains relatively limited in the field of hydrology.

To address the aforementioned challenges, a method is introduced that integrates a physical hydrological model, namely, the SWAT model, with machine learning approaches and involves the use of the SHAP method for model interpretation. In addition, this method not only preserves the physical mechanisms inherent in the SWAT model

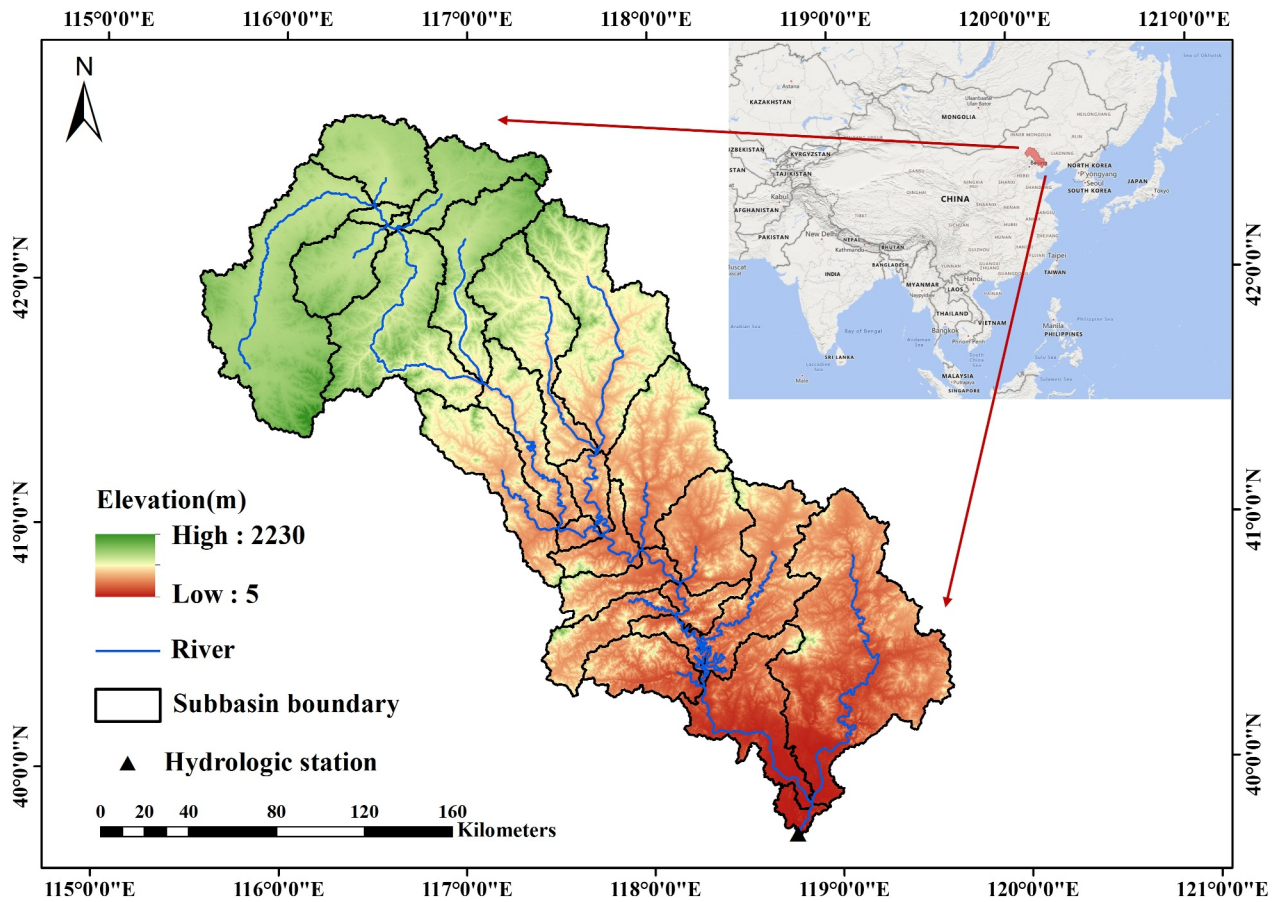


Figure 1. Location of the LRB.

but also leverages the efficiency and interpretability of machine learning models. This study is focused primarily on analyzing the combined effects of climate and land use changes on streamflow, thereby identifying the key controlling factors and their interactions, quantifying the relative contributions of climate and land use changes, and elucidating the complex mechanisms through which they influence streamflow. The findings of this study provide a systematic foundation for water resource management and watershed planning, promote the integrated application of hydrological models and machine learning methods, and enhance the understanding of streamflow processes and their controlling factors.

2. Materials and Methods

2.1. Study Area

The Luan River Basin (LRB) is a subbasin of the Hai River Basin, which is one of China's seven major river basins (Figure 1). With a total length of 888 km, it covers an area of approximately 44,750 square kilometers. This basin, located in a temperate continental monsoon climate zone, experiences hot, rainy summers and cold, dry winters, with average annual precipitation values ranging from 400 to 800 mm and mean temperatures ranging from 1°C to 11°C. The LRB serves as a key water source conservation area for China's Capital Economic Circle and is characterized by a distinct ecotone, where agriculture, pastoral lands, and forests converge. The dominant land use types are cropland, grassland, and forest. Basin streamflow has shown a declining trend, particularly since the implementation of the Grain for Green project, which aims to convert cropland into forestland and grassland (Gao et al., 2023; Wu et al., 2020; Xu et al., 2020). As a result, the pressure on water resources has increased.

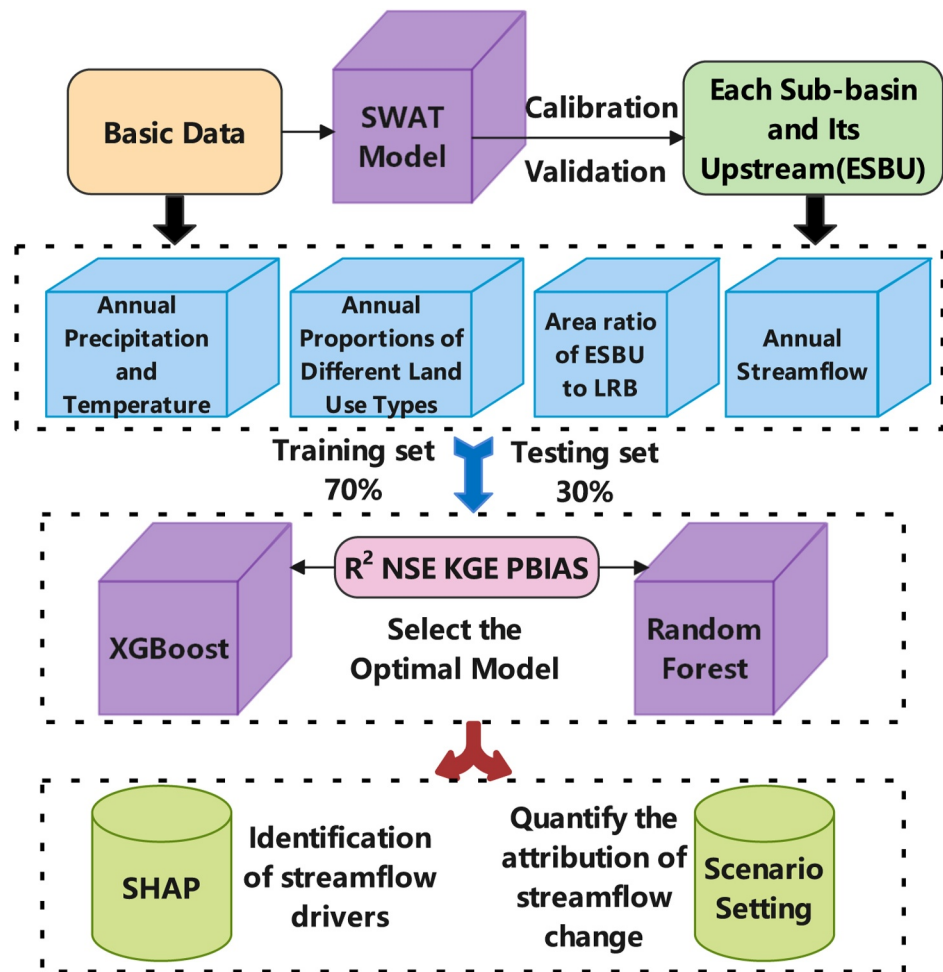


Figure 2. Overall framework of this study.

2.2. Overall Framework and Data

The overall framework of this study is shown in Figure 2. A SWAT model was constructed using foundational data from the study area, including meteorological, hydrological, digital elevation model (DEM), soil, and land use data (refer to Table 1 for details), and it was subsequently calibrated and validated. The calibrated SWAT model was then used to generate streamflow outputs for subbasins. As shown in Figure 3, these outputs represented the combined streamflow of each subbasin and upstream area (ESBU) rather than the streamflow of

Table 1
Data for SWAT Modeling

Data type	Data source	Data description
Digital elevation model data	Version 3 of the ASTER Global Digital Elevation Model (https://data.nasa.gov/dataset/aster-global-digital-elevation-model-v003-7ede1)	30-m-resolution data
Soil data	Harmonized World Soil Database v2.0 (https://www.fao.org/soils-portal/data-hub/en/)	1-km-resolution data
Land use data	Annual land cover product of China (https://zenodo.org/records/8176941)	30-m-resolution data from 1985 to 2020
Meteorological data	Climate Change Research Center (https://ccrc.iap.ac.cn/resource/detail?id=228)	Daily scale data from 1980 to 2020
Streamflow data	Haihe River Water Conservancy Commission, Ministry of Water Resources (MWR) (http://www.hwcc.gov.cn/)	Reconstruction of monthly streamflow data from 1980 to 2020

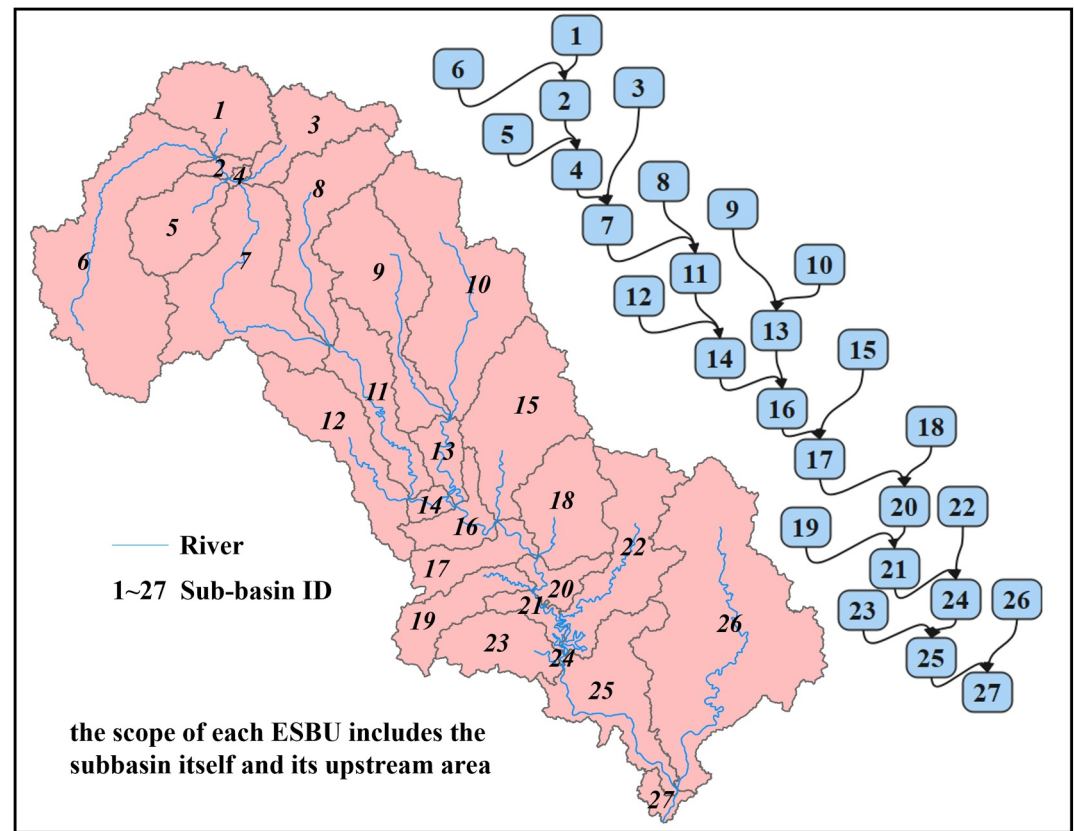


Figure 3. Diagram of the subbasins.

Table 2
Feature Variables and Their Specific Information

Feature variables	Specific definition (annually)
Precipitation	Precipitation within each ESBU
Temperature	Average temperature within each ESBU
Cropland	Proportion of cropland area within each ESBU
Forest	Proportion of forest area within each ESBU
Shrub	Proportion of shrubland area within each ESBU
Grassland	Proportion of grassland area within each ESBU
Water	Proportion of water body area within each ESBU
Barren	Proportion of barren land area within each ESBU
Wetland	Proportion of wetland area within each ESBU
Impervious	Proportion of impervious surface area within each ESBU
Area	Proportion of each ESBU area within the LRB

the subbasin alone. Given the significant variations in streamflow conditions, meteorological conditions, and land use types within the watershed (with relatively constant soil conditions and topography), the annual ESBU streamflow was used as the target variable. The feature variables (Table 2), including the annual precipitation, average temperature, and area proportions of the different land use types (cropland, forest, shrub, grassland, water, barren, wetland, and impervious) and the proportion of the ESBU area within the LRB were employed to develop two machine learning models: XGBoost and RF models. To construct these machine learning models, 70% of the data from each ESBU were used for training, whereas the remaining 30% were reserved for validation. This data allocation method ensured that the models better reflected the information from each ESBU, thereby increasing reliability. The performance characteristics of the two machine learning models were assessed using indicators such as the coefficient of determination (R^2), Nash–Sutcliffe efficiency (NSE), Kling–Gupta efficiency (KGE), and percent bias (PBIAS), and the best-performing model was selected. SHAP analysis, combined with various scenario simulations (Figure 4), was then applied to identify the main factors driving streamflow variations and quantify their relative contributions.

2.3. SWAT Model

The SWAT model, developed by the United States Department of Agriculture (USDA), is a distributed hydrological model with a physical basis. The aim of this model is to simulate the hydrological cycle on the basis of the water balance equation:

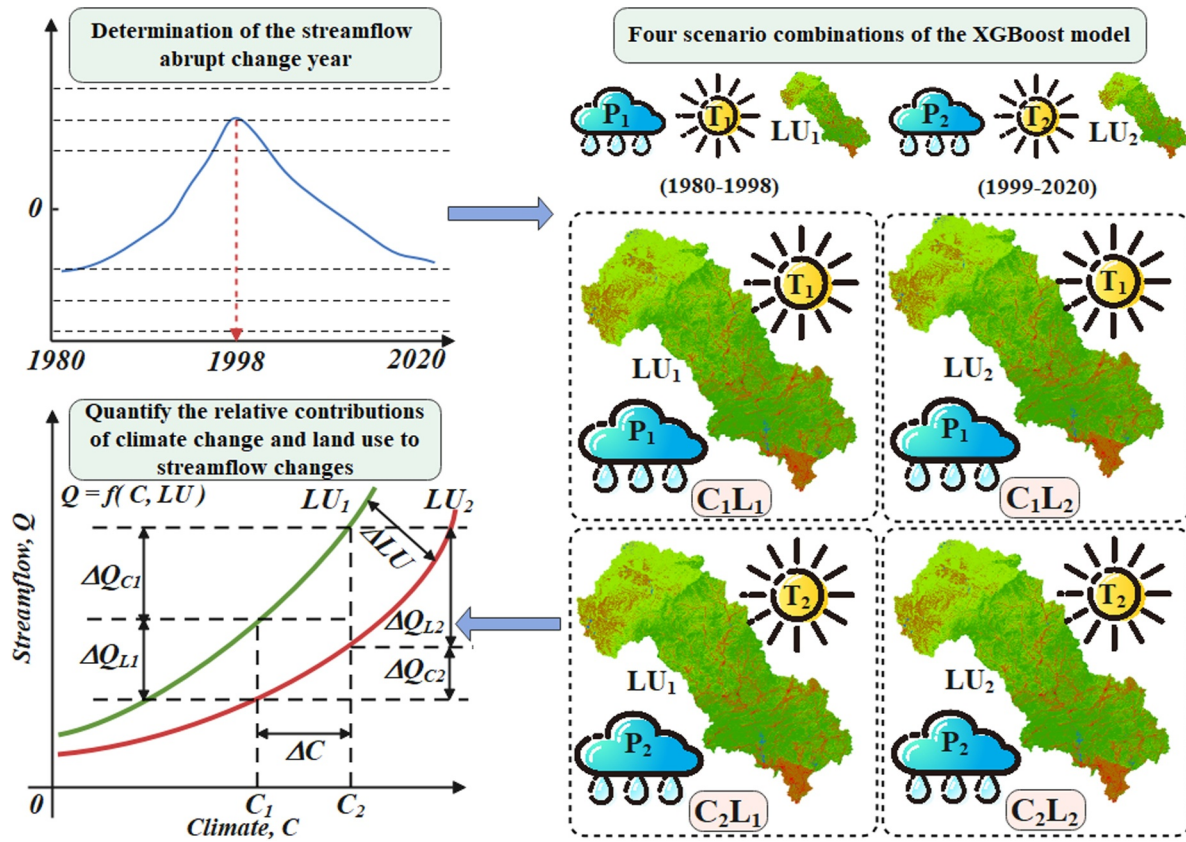


Figure 4. Schematic of the scenario setting and quantitative analysis processes.

$$SW_t = SW_0 + \sum_{i=1}^t (R_{day} - Q_{surf} - E_a - W_{seep} - Q_{gw}) \quad (1)$$

where SW_t denotes the soil water content at the end of period t (mm), SW_0 denotes the initial soil water content (mm), R_{day} denotes the daily precipitation on day i (mm), Q_{surf} denotes the surface streamflow on day i (mm), E_a denotes the actual evapotranspiration on day i (mm), W_{seep} denotes the seepage to the deep soil layer on day i (mm), and Q_{gw} denotes the groundwater return flow on day i (mm).

The SWAT model was calibrated and validated using the Sequential Uncertainty Fitting v2 (SUFI-2) algorithm in SWAT-CUP, as this method is widely recognized for its ability to optimize model parameters and conduct uncertainty analysis (Ding et al., 2024; Zhao et al., 2018). A total of 16 runoff parameters with high sensitivity were selected as calibration parameters for the LRB, with CN2 having the greatest impact (Table 3).

The SWAT model provided the average inflow (FLOW_IN, m^3/s) and outflow (FLOW_OUT, m^3/s) to the river channel at a given time step. Importantly, the FLOW_OUT value was the streamflow of a subbasin and its upstream areas. For example, the FLOW_OUT value of subbasin 2 was the average flow of subbasin 2 and its upstream areas (subbasins 1 and 6), whereas the FLOW_OUT value of subbasin 3 was the streamflow within its boundaries (Figure 3). In this study, the calibrated and validated SWAT model was used to generate the FLOW_OUT value of each ESB, which was then utilized in the subsequent analysis.

2.4. Machine Learning Methods

In this study, two machine learning methods, namely, the XGBoost and RF models, were adopted, both of which were ensemble learning methods (Zhang & Ma, 2012). The annual FLOW_OUT value of each ESB was used as the target variable, while the feature variables included the annual precipitation, average temperature, area proportions of the different land use types, and proportion of the ESB area relative to that of the entire

Table 3
SWAT Model Parameter Selection and Calibration Results

Parameter	Description	Range	Value
CN2	SCS runoff curve number f	(−0.5, 0.5)	0.099919
SOL_BD	Moist bulk density	(−0.5, 0.5)	0.041534
SOL_AWC	Available water capacity of the soil layer	(−0.5, 0.5)	0.079305
SURLAG	Surface runoff lag time	(0.05, 24)	17.482698
SOL_K	Saturated hydraulic conductivity	(−0.5, 0.5)	−0.289509
HRU_SLP	Average slope steepness	(0, 1)	0.279583
SLSUBBSN	Average slope length	(10, 150)	56.821812
RCHRG_DP	Deep aquifer percolation fraction	(0, 1)	0.404313
ESCO	Soil evaporation compensation factor	(0, 1)	0.609442
CANMX	Maximum canopy storage	(0, 100)	12.482609
ALPHA_BF	Baseflow alpha factor (days)	(0, 1)	0.600068
EPCO	Plant uptake compensation factor	(0, 1)	0.515179
GW_REVAP	Groundwater “revap” coefficient	(0.02, 0.2)	0.066258
GW_DELAY	Groundwater delay (days)	(0, 500)	339.624908
BIOMIX	Biological mixing efficiency	(−0.5, 0.5)	−0.273913
REVAPMN	Threshold depth of water in the shallow aquifer for “revap” to occur (mm)	(0, 500)	441.635864

watershed. The land use types were classified into seven categories: cropland, forestland, shrubland, grassland, water bodies, barren land, impervious surfaces, and wetland.

The XGBoost model is an efficient implementation of gradient boosting decision trees (GBDTs) that provides enhanced prediction performance by sequentially adding new trees to fit the residuals of previous iterations (Chen & Guestrin, 2016). This model leverages second-order derivative information to optimize the loss function and incorporates a regularization term to control model complexity, thereby preventing overfitting. The objective function could be expressed as follows:

$$Obj^{(t)} = \sum_{i=1}^n \left[l(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)) \right] + \Omega(f_t) \quad (2)$$

where $l(y_i, \hat{y}_i)$ is the loss function, which captures the difference between the true value y_i and the predicted value $\hat{y}_i$; $\hat{y}_i^{(t-1)}$ is the prediction at the $t-1$ -th iteration; $f_t(x_i)$ is the prediction of the t -th tree for sample x_i ; and $\Omega(f_t)$ is the regularization term for the t -th tree.

The RF model provides increased prediction performance via the construction of multiple independent decision trees and the aggregation of their results (Breiman, 2001). This model introduces randomness in both the samples and features, thus ensuring the diversity of each tree and reducing the overall variance in the model. For regression problems, the prediction of the RF model is the average of the predictions from all decision trees, which could be expressed as follows:

$$\hat{y} = \frac{1}{N} \sum_{i=1}^N f_i(x) \quad (3)$$

where $\hat{y}$ is the predicted value of the model for input x , N is the number of decision trees, and $f_i(x)$ is the prediction of the i th decision tree. This ensemble learning method effectively reduced overfitting issues that resulted from the use of a single decision tree by averaging the predictions from multiple trees, thus enhancing the generalizability of the model.

2.5. Model Evaluation Metrics

To evaluate the performance characteristics of the SWAT, XGBoost, and RF models, four evaluation metrics were selected: R^2 , NSE, KGE, and PBIAS. R^2 primarily captures linear correlation, NSE emphasizes overall error analysis, KGE provides a comprehensive assessment, and PBIAS directly reflects the systematic bias of the model. The specific calculation equations could be expressed as follows:

$$R^2 = \frac{[\sum_{i=1}^n (O_i - \bar{O})(P_i - \bar{P})]^2}{\sum_{i=1}^n (O_i - \bar{O})^2 \sum_{i=1}^n (P_i - \bar{P})^2} \quad (4)$$

$$NSE = 1 - \frac{\sum_{i=1}^n (O_i - P_i)^2}{\sum_{i=1}^n (O_i - \bar{O})^2} \quad (5)$$

$$KGE = 1 - \sqrt{(r-1)^2 + (\alpha-1)^2 + (\beta-1)^2} \quad (6)$$

$$PBIAS = 100 \times \frac{\sum_{i=1}^n (P_i - O_i)}{\sum_{i=1}^n O_i} \quad (7)$$

where O_i is the observed value, P_i is the simulated value, $\bar{O}$ is the mean of the observed values, $\bar{P}$ is the mean of the simulated values, r is the correlation coefficient between the simulated and observed values, $\alpha = \sigma_P/\sigma_O$ is the ratio of the standard deviations (variability), and $\beta = \mu_P/\mu_O$ is the ratio of the means (bias).

2.6. SHAP

The SHAP values are based on Shapley value theory from game theory, thereby providing a theoretically rigorous and widely applicable method for interpreting machine learning model predictions (Lundberg, 2017). SHAP values fairly reflect the contribution of each feature to the model prediction characteristics by calculating the marginal contribution of each feature under all possible feature combinations. The SHAP value could be calculated as follows:

$$SHAP(x_i) = \sum_{S \subseteq \{x_1, \dots, x_n\} \setminus \{x_i\}} \frac{|S|!(n-|S|-1)!}{n!} [f(S \cup \{x_i\}) - f(S)] \quad (8)$$

where S is any subset of features that does not include feature x_i , $f(S)$ is the model output when only the features in S are included, and $f(S \cup \{x_i\})$ is the model output when x_i is added to S . The weight $\frac{|S|!(n-|S|-1)!}{n!}$ ensures fairness and symmetry across all possible feature combinations. The uniqueness of SHAP values lies in their unification of multiple explanation models, thus ensuring the consistency and additivity of the explanation results. In the context of SHAP values, positive and negative changes represent the contributions of individual features to the model's prediction relative to the baseline. Positive SHAP values indicate that the feature increases the predicted outcome, whereas negative SHAP values suggest a reduction in the prediction. These values are an intuitive method for understanding the mechanism by which each feature impacts the model's prediction in specific instances. Specifically, relative to the baseline, positive SHAP values imply that the feature's value pushes the prediction higher, whereas negative SHAP values indicate that the feature's value pushes the prediction lower.

2.7. Scenario Setting

The fixed-change method is commonly employed to distinguish and quantify the independent impacts of climate and land use changes on variations in watershed streamflow and water balance components (Sharma et al., 2022). The Pettitt test was first applied to streamflow time series data to detect breakpoints (Figure 4). The study period was then divided into baseline and post-change periods. In this study, a breakpoint was determined in 1998 on the basis of the detection results, which allowed for the analysis of changes and the independent effects of climate and land use changes. The climate and land use data for the baseline period (1982–1998) were denoted as C_1 and LU_1 , respectively, whereas the climate and land use data for the post-change period (1999–2020) were denoted as C_2 and LU_2 , respectively.

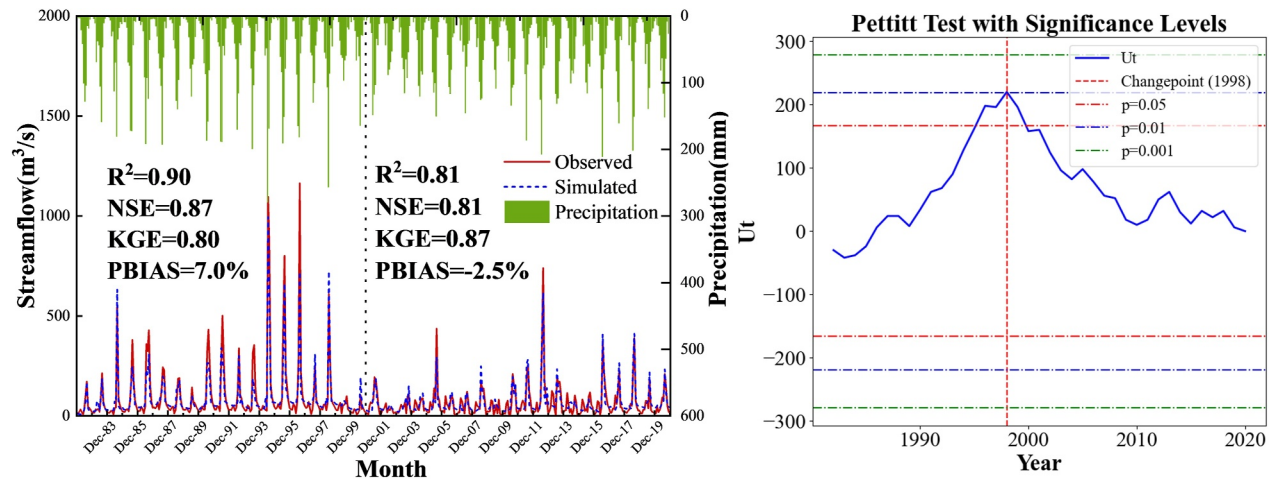


Figure 5. SWAT model simulation performance (left) and streamflow Pettitt test results (right).. (The dashed lines represent significance levels: red for $p = 0.05$, blue for $p = 0.01$, and green for $p = 0.001$. The red vertical line marks the changepoint year).

On the basis of the theoretical framework proposed in previous studies (Sharma et al., 2022), four climate and land use combinations were derived: L_1C_1 , L_1C_2 , L_2C_1 , and L_2C_2 . These combinations were then simulated using the selected machine learning model. The outputs for these combinations were analyzed to quantify the contributions of land use change (LUC) and climate change (CC) to streamflow variation. The total change in streamflow (ΔQ) was driven by both climate (ΔQ_C) and land use (ΔQ_L), and the specific calculation equations could be expressed as follows:

$$\Delta Q = \Delta Q_C + \Delta Q_L \quad (9)$$

$$\Delta Q_C = \frac{1}{2}[(Q_{L_1}^{C_2} - Q_{L_1}^{C_1}) + (Q_{L_2}^{C_2} - Q_{L_2}^{C_1})] \quad (10)$$

$$\Delta Q_L = \frac{1}{2}[(Q_{L_2}^{C_1} - Q_{L_1}^{C_1}) + (Q_{L_2}^{C_2} - Q_{L_1}^{C_2})] \quad (11)$$

3. Results

3.1. Meteorological, Hydrological, and Land Use Changes

The SWAT model effectively simulated streamflow variations in the LRB. The simulation time step was set to a monthly scale, with a warm-up period from 1980 to 1982 (Figure 5). The model performance was assessed using evaluation metrics, with R^2 , NSE, and KGE values greater than 0.6 indicating good performance and PBIAS values between $\pm 10\%$ and $\pm 15\%$ indicating acceptable performance. During the calibration period (1982–2000) and the validation period (2000–2020), the model achieved R^2 , NSE, and KGE values above 0.8, with PBIAS values of 7.0% and -2.5% , respectively. All the evaluation metrics were satisfactory, indicating reliable FLOW_OUT values for each ESB from the SWAT model output. Between 1980 and 2020, the streamflow in the LRB exhibited a downward trend. Through the use of Pettitt's test for change-point analysis, the year 1998 was identified as exhibiting a significant change in streamflow. Prior to 1998 (the baseline period), the multiyear average streamflow was $104.18 \text{ m}^3/\text{s}$, whereas after 1998 (the post-change period), the value decreased to $63.97 \text{ m}^3/\text{s}$, representing a decline of $40.21 \text{ m}^3/\text{s}$. Additionally, the multiyear average annual precipitation decreased from 531.31 mm during the baseline period to 500.57 mm during the post-change period, reflecting a reduction of 30.74 mm . Moreover, the annual average temperature slightly increased from 5.32°C to 5.82°C with no significant fluctuations.

The significant reduction in streamflow in the basin was influenced not only by decreased precipitation but also by substantial land use changes. The primary land use types in the LRB, namely, cropland, forest, and grassland, collectively accounted for more than 95% of the total area (Figure 6). The main changes included a continuous

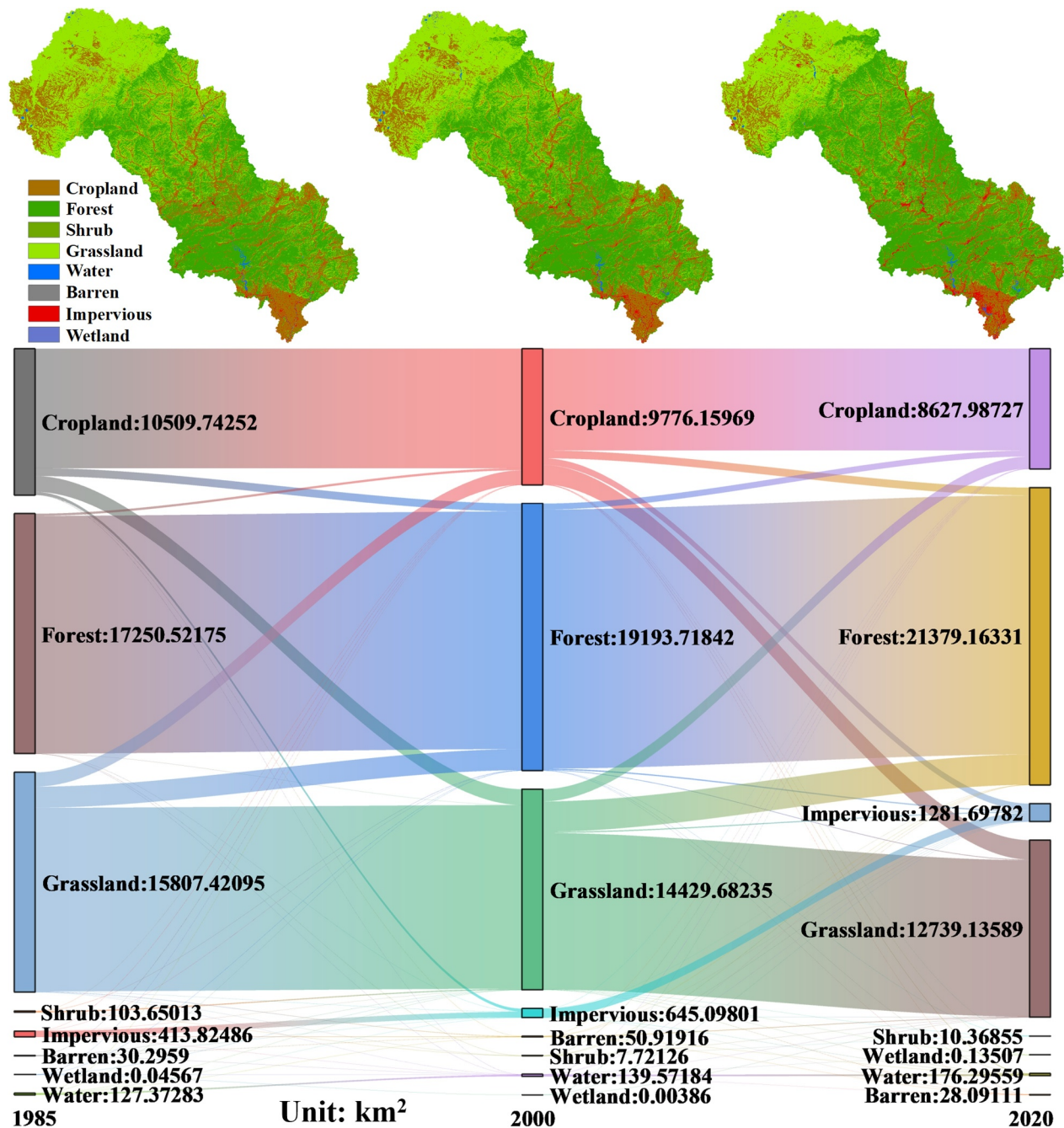


Figure 6. Land use type changes in the LRB.

reduction in cropland and grassland areas, while the forest area steadily increased, likely due to the implementation of the Grain for Green program. The cropland area decreased from 10,509.74 km² in 1985 to 9,776.16 km² in 2000 and further decreased to 8,627.99 km² by 2020, whereas the grassland area exhibited a greater reduction. Cropland was primarily converted into forest, grassland, and impervious surfaces. Before 2000, grassland was mainly converted into forest and cropland, whereas after 2000, it was primarily converted into forest. As a result, the forest area continued to increase, namely, from 17,250.52 km² in 1985 to 21,379.16 km² in 2020. Additionally, urbanization caused the area of impervious surfaces to increase by 867.87 km² by 2020 compared with that in 1985.

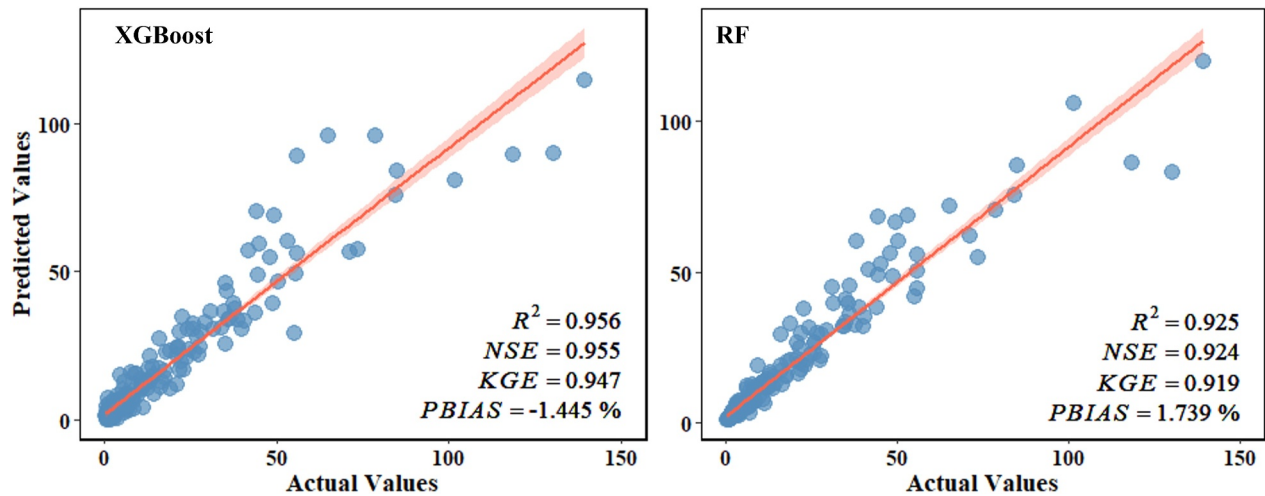


Figure 7. Performance comparison of the XGBoost and RF models.

3.2. Machine Learning Model Selection

On the basis of the streamflow outputs of the calibrated and validated SWAT model combined with relevant climate and land use data, both the XGBoost and RF models were trained. Both models exhibited notable streamflow simulation and prediction capabilities, with the XGBoost model performing better (Figure 7). The XGBoost algorithm outperformed the RF model in all the evaluation metrics. The R^2 values of the XGBoost and RF models were 0.956 and 0.925, respectively, whereas the NSE values were 0.955 for the XGBoost model and 0.924 for the RF model. The KGE was higher for the XGBoost model at 0.947. Notably, the PBIAS values of the XGBoost and RF models were -1.445% and 1.739% , respectively, indicating that, compared with the observations, the XGBoost model underestimated the streamflow, whereas the RF model overestimated it. Both models performed better when simulating smaller values, but their performance decreased when simulating larger values. This was due to the lower frequency of larger values, which resulted in fewer data points for model training. Considering the overall performance, the XGBoost model was selected for simulating streamflow in the LRB.

To validate the performance of the XGBoost model in simulating the annual streamflow in the LRB, we compared its results with those of the SWAT model using observed data (Figure 8). The R^2 and NSE values for the SWAT simulation results were 0.827 and 0.890, respectively, whereas the KGE and PBIAS values were 0.795% and

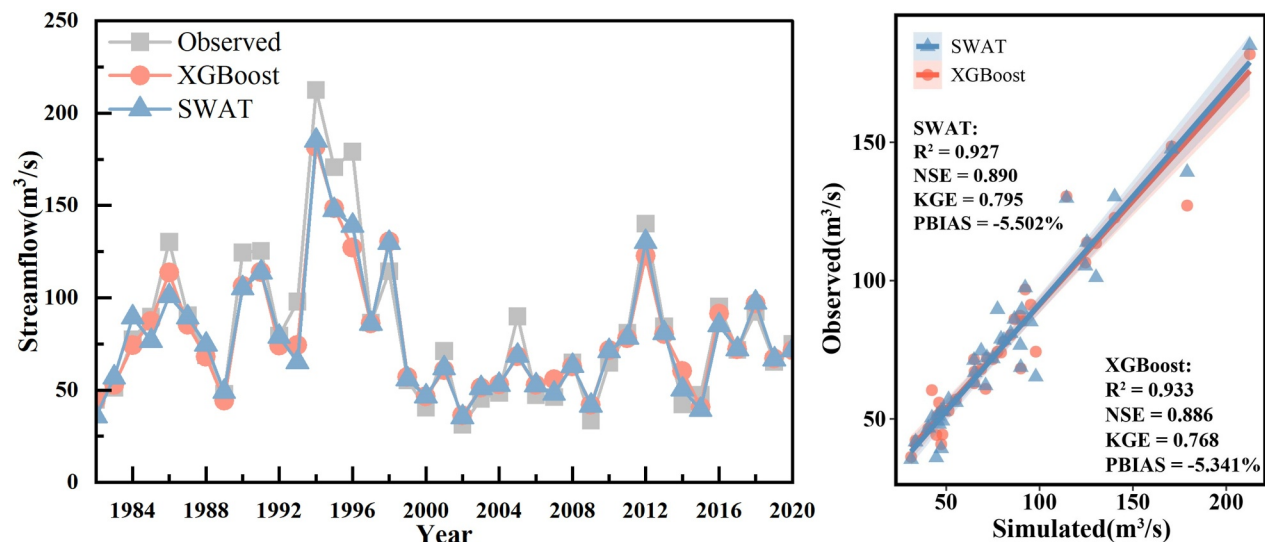


Figure 8. Comparison of the streamflow simulations between the XGBoost and SWAT models.

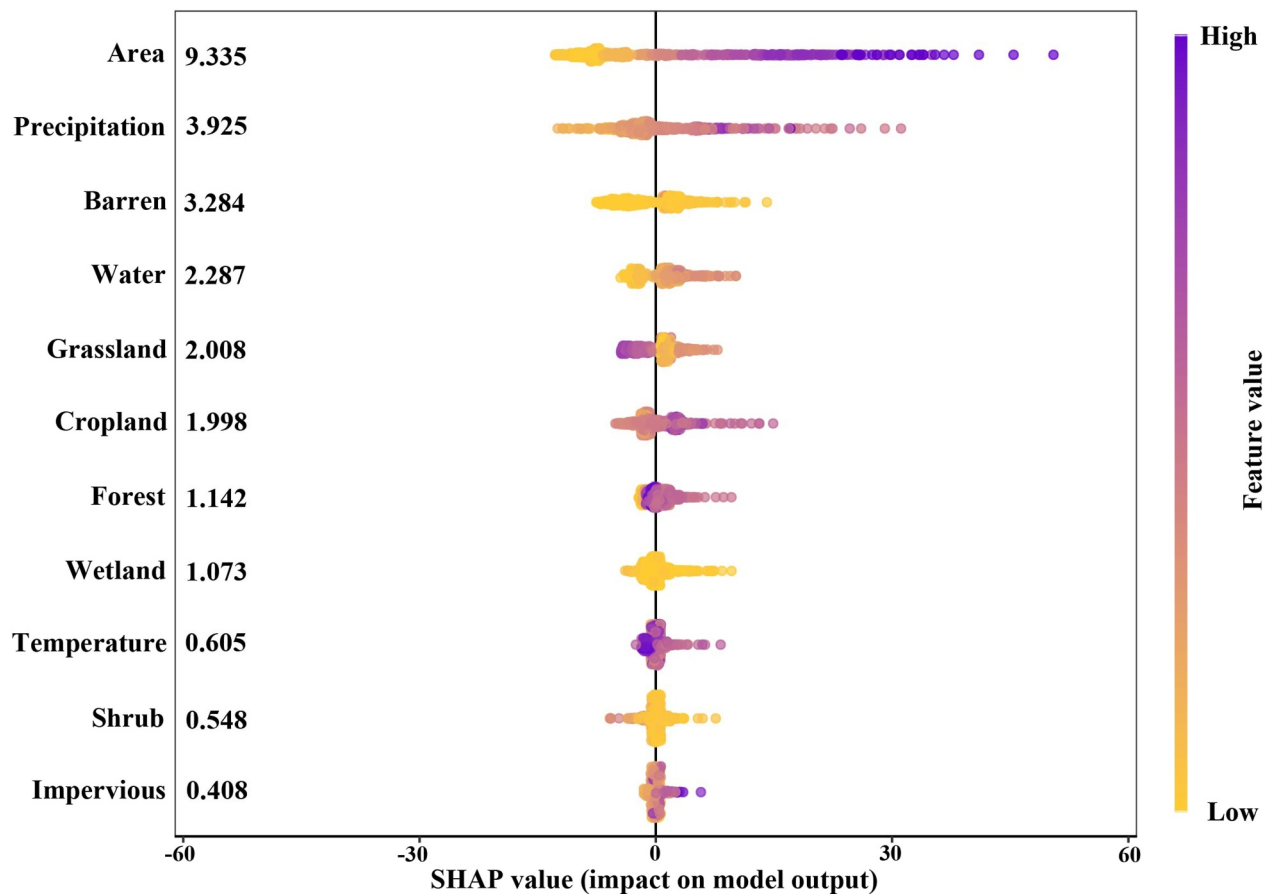


Figure 9. SHAP summary plot of the contributions of the different feature variables to streamflow changes.

−5.502%, respectively. The XGBoost model demonstrated satisfactory simulation ability, with NSE and KGE values of 0.886 and 0.768, respectively, which were slightly lower than those of the SWAT model. However, the R^2 and PBIAS values were superior to those of the SWAT model. Overall, the traditional hydrological SWAT model could effectively simulate the annual streamflow in the LRB, whereas the machine learning XGBoost model, trained on SWAT model outputs, yielded satisfactory simulation results. This finding enabled us to apply the XGBoost model, combined with different scenario settings, to quantitatively calculate the relative contributions of climate and land use changes to streamflow variations. The combined use of the SWAT and XGBoost models, while ensuring the accuracy of the results, significantly increased the computational efficiency and avoided the need for repetitive modeling and parameter adjustments in the SWAT model.

3.3. Controlling Factors of Streamflow Variability

The construction of an accurate streamflow simulation and prediction model was considered crucial, as a more precise model could better capture the relationship between streamflow and its controlling factors. Equally important was the ability to interpret the modeling results effectively. In this study, we applied the SHAP method to interpret the XGBoost model results, allowing us to determine the contribution of each controlling factor to streamflow variability.

3.3.1. Total Effects of the Controlling Factors

To analyze the impacts of climate and land use changes on streamflow variability, the SHAP method was employed. Figure 9 shows the ranking of the feature variables by importance, with the area identified as the most significant factor. With increasing ESBV area, the SHAP value increased, suggesting that streamflow in the basin accordingly increased, conforming with fundamental hydrological principles. In larger basins, more precipitation

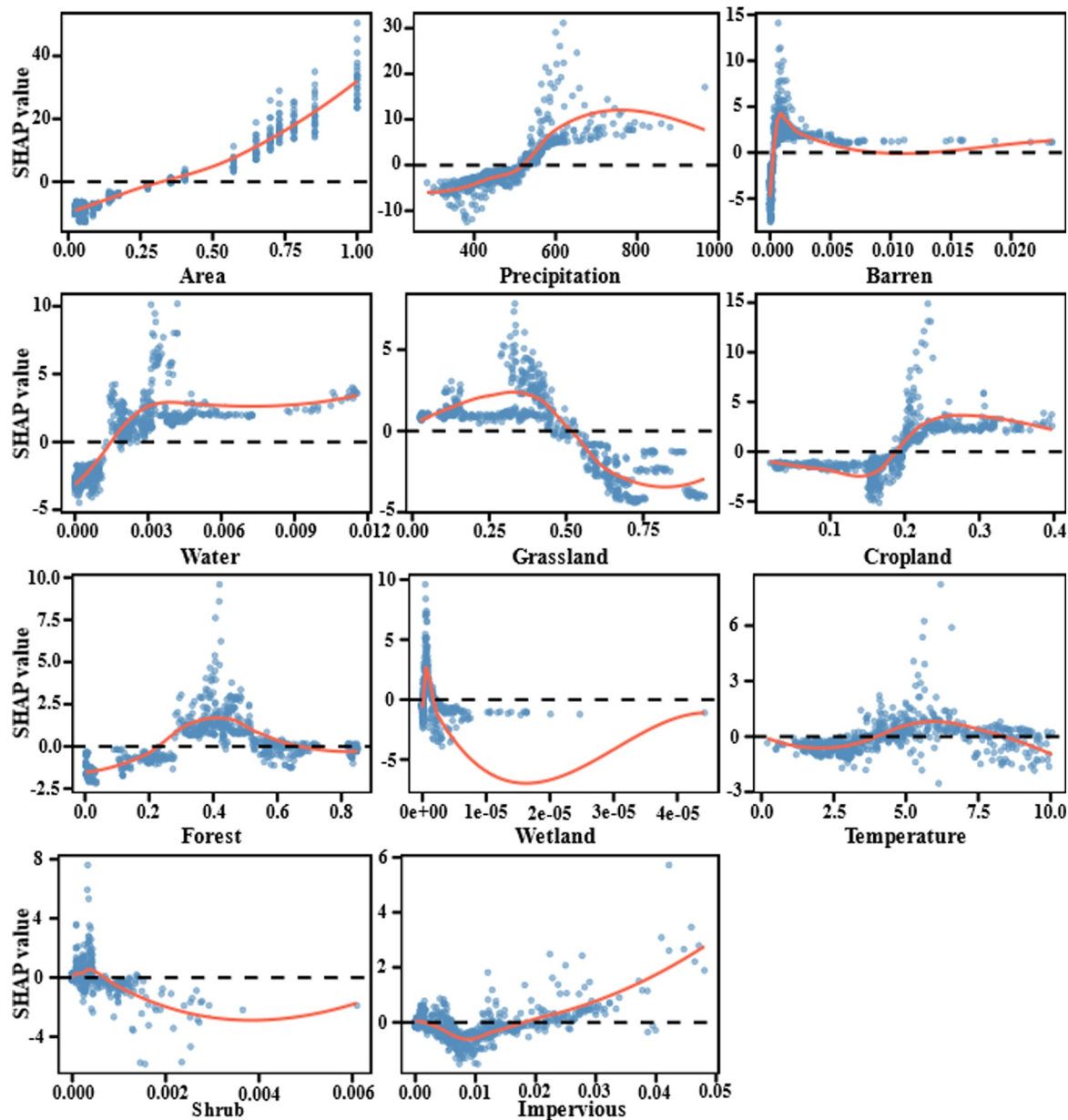


Figure 10. SHAP dependency plot of the XGBoost model. Positive (negative) SHAP values indicated a positive (negative) contribution of the feature variable to the target variable.

and hydrological inputs can accumulate, leading to higher streamflow. Precipitation was the second most important feature variable, with the SHAP value increasing with increasing precipitation. When the precipitation level exceeded 550 mm, it positively contributed to streamflow (Figure 10). In comparison, temperature had a much smaller influence on streamflow, as indicated by its lower SHAP value. The order of importance of the different land use types was as follows: barren land, water bodies, grassland, cropland, forest, wetland, and impervious surfaces. Although cropland, grassland, and forest accounted for the majority of the total watershed area and presented relatively stable hydrological characteristics, smaller land cover types, such as barren land and water bodies, could impact streamflow more notably than larger types because of their unique hydrological properties. Barren areas provided relatively low water retention capacity, leading to increased streamflow. Conversely, water bodies could play a critical role in regulating streamflow by directly affecting the storage and release of water. This phenomenon highlighted the complexity of the hydrological system of the watershed, where

even land cover types occupying a low proportion of the area could fulfill a crucial role in shaping streamflow dynamics owing to their unique influence mechanisms.

The impacts of the different land use types on the SHAP values were nonlinear with notable differences, and threshold effects were observed (Figure 10). Given a barren land proportion of 0, the SHAP value was negative, indicating a negative impact on streamflow. As the barren land proportion slightly increased, the SHAP value increased rapidly, resulting in a significant positive impact on streamflow. However, as the barren land proportion continued to increase, the positive impact decreased sharply and eventually approached 0. The impact of water bodies on streamflow exhibited significant nonlinearity. When the proportion of water bodies was low ($<0.15\%$), streamflow was negatively affected. However, as the proportion of water bodies increased, its regulating effect increased accordingly, causing the SHAP value to increase sharply. As the proportion of water bodies continued to increase ($>0.5\%$), the impact gradually stabilized. Consequently, the rate of increase in the SHAP values decreased and eventually stabilized. The correlation between grassland and streamflow was generally negative, with the SHAP value decreasing with increasing grassland proportion. When the grassland proportion exceeded 50%, it started to negatively impact streamflow. In contrast, the effect of cropland was the opposite, with the SHAP value increasing with increasing cropland proportion. When the cropland proportion exceeded 20%, it started to positively affect streamflow. The impact of forestland on streamflow followed a parabolic pattern. The SHAP value first increased but then decreased with increasing forest proportion. When the forestland proportion varied between 30% and 50%, it positively affected streamflow, peaking at approximately 40%. However, as the forestland proportion continued to increase, the SHAP value became negative, which agreed with the findings of many studies suggesting that excessive afforestation could lead to a reduction in streamflow. The wetland and shrubland proportions were relatively low, and they exerted a positive influence only at very low proportions, with a mostly negative impact observed as their area increased. The overall impact of impervious surfaces on streamflow was positive, with SHAP values increasing with increasing proportions of impervious surfaces. The positive effect of impervious surfaces on streamflow mainly resulted from their disruption of water infiltration and storage processes, causing more precipitation to be converted into streamflow. We observed that SHAP values increased with the proportion of area, and when the area exceeded 40%, the SHAP value became positive, indicating a positive contribution to streamflow. This finding suggested a threshold effect, where the positive impact of area on streamflow became more pronounced as the ESBU area increased.

3.3.2. Interaction Effects of the Controlling Factors

When analyzing the impacts of individual features on the target variable through SHAP dependence plots, we observed vertical diffusion, which was caused by the influences of other features. Therefore, to better understand the interaction effects of the controlling factors on streamflow variation, we applied interaction analysis. Owing to the high proportions of grassland, cropland, forestland, and impervious surfaces in the watershed area and the significant changes in these land use types, we comprehensively analyzed the interaction effects among these major land use types and their interactions with precipitation. This finding could provide a comprehensive understanding of the complex mechanisms underlying watershed streamflow processes and their controlling factors.

The interaction effects on streamflow among the different land use types differed notably (Figure 11). The interaction between cropland and forestland showed a nonlinear change in the SHAP values, which generally increased under fluctuations with increasing cropland proportion. When the cropland proportion was low ($<10\%$), the forest proportion was also low, and the SHAP interaction value was negative, indicating a negative contribution. As the cropland proportion increased to a value between 10% and 23%, the forestland proportion increased, and the interaction effect resulted in both positive and negative contributions. When the cropland proportion increased further and the forestland proportion was low, the interaction effect indicated a positive contribution. However, when the forestland proportion was high, the interaction effect became negative. The interaction between cropland and grassland showed a nonlinear change in the SHAP values, which generally decreased under fluctuations as the proportion of cropland increased. When the cropland proportion was less than 8%, the grassland proportion was relatively high, and the SHAP interaction value was positive, indicating a positive contribution. As the proportion of cropland increased to a value between 8% and 20%, the interaction effect reflected a positive contribution when the proportion of grassland was high but a negative contribution when the proportion of grassland was low. With a further increase in the proportion of cropland, the interaction effect was positive when the proportion of grassland was low but negative when the proportion of grassland was high. The SHAP interaction value between cropland and impervious surfaces showed a fluctuating upward trend

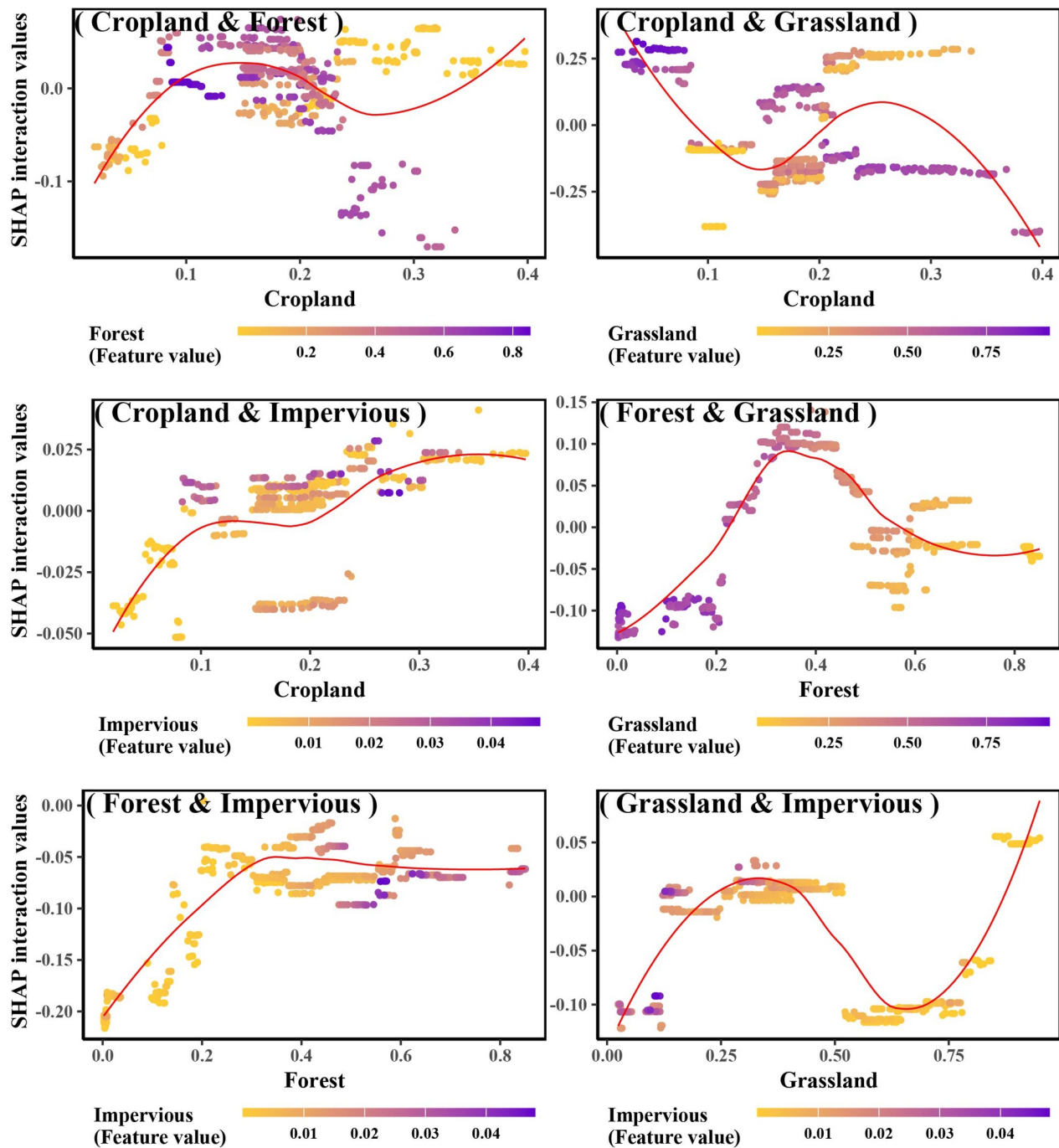


Figure 11. Interaction effect plots for the different land use types.

with increasing cropland proportion. When the proportions of both land use types were relatively high or when the proportion of one land use type was particularly high, the SHAP value was positive, indicating a positive contribution. The SHAP interaction value between cropland and forestland first increased but then decreased with increasing forestland proportion. When the forestland proportion was low (<25%) and the grassland proportion was relatively high, the SHAP interaction value was negative, indicating a negative contribution to streamflow. As the forestland proportion increased to a value between 25% and 45% and the grassland proportion was moderate, the interaction effect was positive. However, when the forestland proportion further increased and the grassland proportion decreased, the interaction effect became negative. Overall, the interaction effect between

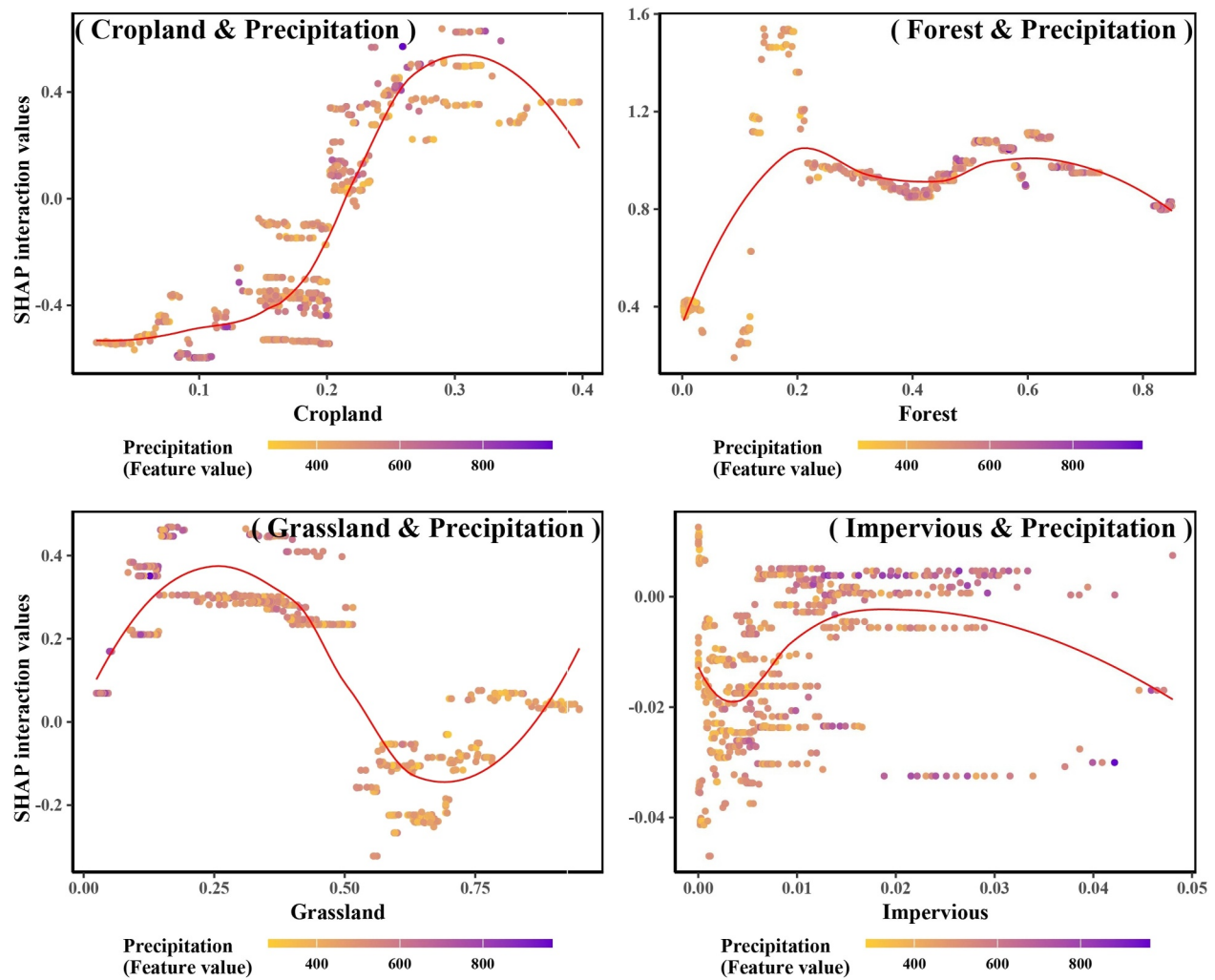


Figure 12. Interaction effect plots between precipitation and the different land use types.

forestland and grassland was positive when their proportions were balanced and negative when the proportions were imbalanced (i.e., the proportion of one was very high and that of the other was very low). The SHAP interaction value between forestland and impervious surfaces was always negative, suggesting an overall trend of initial increase and subsequent stabilization with increasing forestland proportion. Under low proportions of forestland and impervious surfaces (<30%), the SHAP interaction value gradually increased. However, as the proportions of both land use types continued to increase, the SHAP value stabilized, and the interaction effect consistently generated a negative contribution to streamflow. The SHAP interaction value between grassland and impervious surfaces exhibited a nonlinear pattern. When the proportion of grassland was low (<20%) and the proportion of impervious surfaces was high, the SHAP interaction value was negative. Interestingly, even when the proportion of grassland was high (>50%) and the proportion of impervious surfaces was low, the SHAP interaction value remained mostly negative, thereby negatively contributing to streamflow.

The interaction effects between precipitation and cropland, forestland, grassland, and impervious surfaces varied notably, exhibiting distinct nonlinear characteristics and response patterns (Figure 12). These land use types exhibited unique mechanisms in their interactions with precipitation, and the variations in land use types impacted streamflow more notably than did those in precipitation, highlighting the dominant role of land use types in determining interaction effects.

The interaction between precipitation and cropland revealed that precipitation had a relatively limited effect on the SHAP interaction value, whereas the cropland proportion had a more significant impact. Specifically, when

the cropland proportion was low, the SHAP interaction values were negative, indicating that the interaction between precipitation and cropland negatively contributed to streamflow, regardless of the precipitation level. However, with increasing cropland proportion, the SHAP interaction value gradually increased, indicating a positive effect. When the cropland proportion reached a critical threshold (approximately 20%–22%), a positive influence of cropland began to emerge, even at lower precipitation levels, thereby effectively increasing streamflow. The SHAP interaction value between precipitation and forestland consistently remained positive, indicating a positive contribution of their interaction to streamflow. When the forestland proportion was less than 20%, the SHAP interaction value increased sharply with increasing forestland proportion, regardless of the precipitation level. As the forestland proportion further increased, the SHAP values slightly fluctuated; specifically, the values initially decreased, subsequently increased, and finally decreased again. In this process, the variation in precipitation was notable, suggesting that precipitation did not dominate the interaction effects. Instead, changes in the forestland proportion exerted a more significant influence on the interaction effects.

The SHAP interaction values between precipitation and grassland first increased, then decreased, and finally increased again. When the proportion of grassland was less than 50%, the SHAP interaction values were positive, indicating that an increase in the grassland area positively contributed to streamflow. However, when the grassland proportion exceeded 50%, the SHAP interaction values became negative, suggesting that an increase in the grassland area negatively impacted streamflow. Importantly, in this process, the precipitation level varied considerably, further emphasizing that changes in the grassland proportion exerted a more significant influence on the interaction effects, whereas the role of precipitation was relatively limited. The SHAP interaction values between precipitation and the proportion of impervious surfaces were typically negative when the proportion of impervious surfaces was low (<1%), indicating a negative contribution of the interaction effect to streamflow. As the proportion of impervious surfaces increased, the SHAP interaction values were mostly positive, indicating a positive contribution to streamflow. However, negative contributions remained, suggesting that the increase in impervious surfaces could trigger complex interaction effects in certain cases. Notably, throughout this process, considerable fluctuations in precipitation levels occurred, further indicating that the interaction between precipitation and the proportion of impervious surfaces was complex.

3.4. Quantifying the Relative Contributions of the Controlling Factors

The interaction effects between precipitation and several major land use types suggested that precipitation did not dominate these interactions. To quantify the impacts of climate and land use changes on streamflow, we conducted a more in-depth analysis based on the framework outlined in Section 2.7.

In the LRB, streamflow mutation occurred in 1998 (Figure 5). On the basis of the four scenario settings, the relative contributions of climate and land use changes to streamflow variation were effectively quantified, represented by $\Delta Q_C/\Delta Q$ and $\Delta Q_L/\Delta Q$, respectively. We first used the trained XGBoost model to simulate streamflow under the four scenarios and then applied Equations 9–11 to calculate ΔQ , ΔQ_C , and ΔQ_L for each ESB (Figure 13). Compared with those during the baseline period, during the postchange period, the streamflow decreased by 0.37 m³/s to 39.61 m³/s (ΔQ), with a reduction due to climate change ranging from 0.06 to 18.01 m³/s (ΔQ_C), whereas land use change caused a greater reduction in streamflow, ranging from 0.29 to 22.06 m³/s (ΔQ_L). For the entire LRB, the ΔQ_C and ΔQ_L values were 17.54 and 22.06 m³/s, respectively.

The relative contributions of climate and land use changes varied across the different ESBs, but in all ESBs, streamflow changes were driven primarily by land use change rather than by climate change. The relative contribution of land use change ranged from 51.72% to 91.81%, whereas the corresponding contribution of climate change ranged from 8.19% to 48.28%. For the entire LRB, the relative contribution of land use change was 55.71%, and that of climate change was 44.27%. This finding indicated that streamflow mutation in the LRB was caused mainly by significant land use changes, although climate change played a facilitating role. As previously analyzed, precipitation played a relatively minor role in determining interactions with major land use types, leading to a greater relative contribution of land use change. The differences in the relative contributions of climate and land use changes across the different ESBs reflected the varying interactions between precipitation and the different land use types.

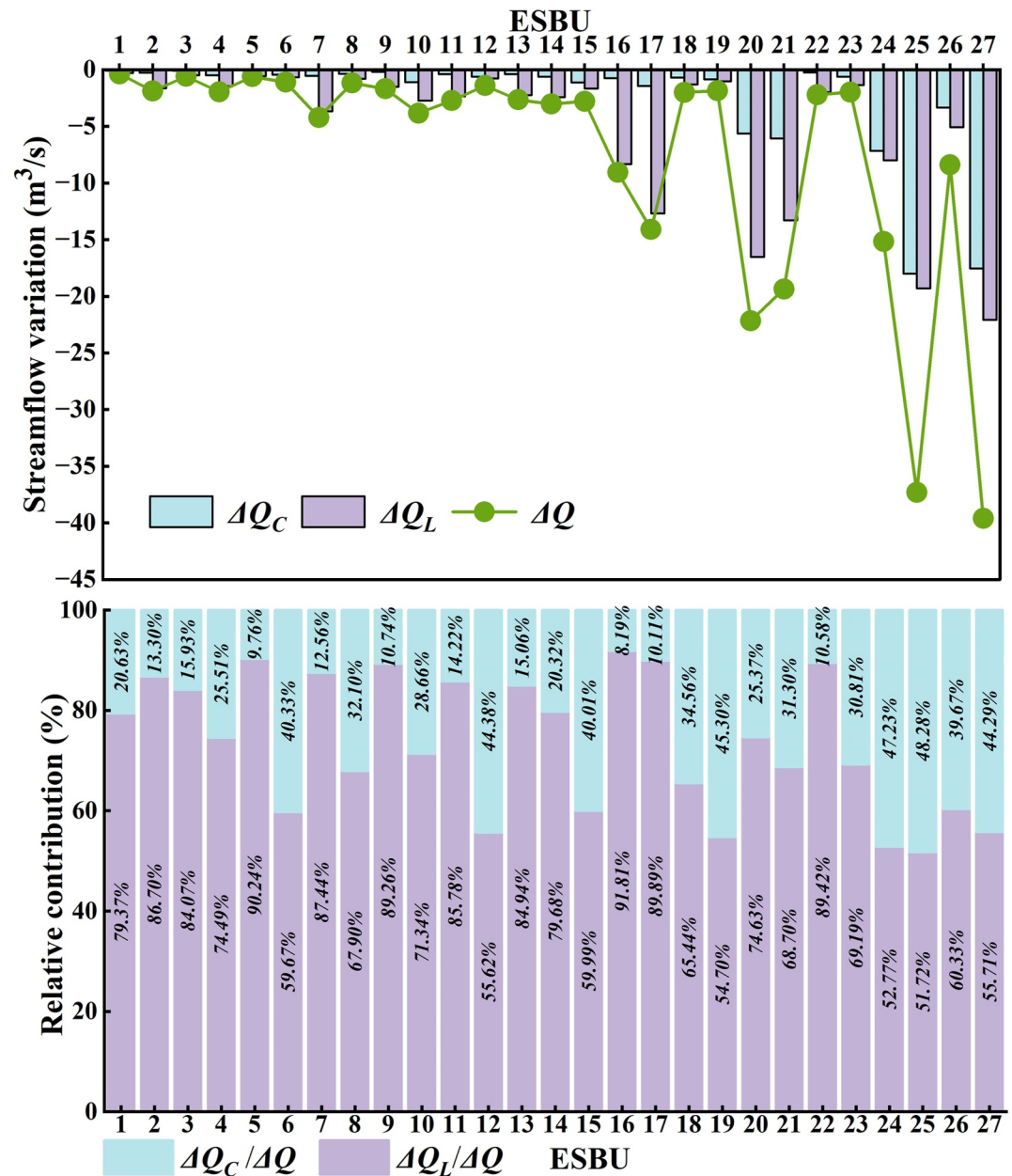


Figure 13. Relative contributions of climate and land use changes to streamflow variation.

4. Discussion

4.1. Applicability of the Research Methods

In hydrological research, accurately simulating streamflow changes and identifying the key controlling factors are fundamental for understanding watershed hydrological processes and effectively managing water resources. In recent years, the combination of physically based hydrological models and machine learning has been recognized as a cutting-edge and promising approach in Earth science research. In this study, the SWAT model is integrated with the XGBoost algorithm, demonstrating the potential of combining these two approaches. Owing to its rich physical mechanisms for simulating streamflow transfer (Arnold et al., 2012), the SWAT model generates streamflow time series data that are inherently linked to hydro-physical processes (Wang et al., 2022). These data not only serve as the external manifestation of hydrological process simulations but also provide physically guided information for training machine learning models. When these data are used to train the XGBoost model,

the resulting model combines the flexibility of data-driven approaches with the theoretical guidance of physical mechanisms, as there are inherent correlations among the variables used for model training (Liang et al., 2020; Wang et al., 2022). By integrating a physical model, the proposed approach enhances the ability of the model to predict streamflow variations accurately. Furthermore, influences from the upstream area on streamflow in each subbasin are incorporated. Instead of treating each subbasin as an independent unit, the concept of the ESB is employed to reflect the interconnectedness of watershed processes comprehensively (Wang et al., 2024). Additionally, in data allocation, 70% of the data from each ESB were used for model training rather than adopting a uniform 70% split to the entire dataset. This approach allows for a realistic representation of real-world conditions, thereby increasing the performance and reliability of the XGBoost model.

This framework has potential applicability in other regions, as the integration of a physically based hydrological model with machine learning offers a versatile approach that can be adapted to various geographic and environmental conditions. The methodology can be extended to other watersheds with similar hydrological characteristics, provided that appropriate data on precipitation, temperature, and land use are available. Moreover, the use of the ESB concept to account for upstream influences on streamflow can be applied to other regions where the spatial interactions between subbasins play a significant role in watershed hydrology. The combination of physical models and machine learning holds promise for enhancing the prediction accuracy of streamflow variations in diverse contexts, contributing to effective water resource management. Additionally, the use of SHAP combined with scenario simulations provides valuable insights into the factors controlling streamflow variation and can be used to quantify the relative contributions of climate and land use changes. This approach further strengthens the generalizability of the methodology by offering a robust framework for understanding how different factors influence streamflow in various regions.

4.2. Interpretability of the Controlling Factors for Streamflow Variation

Although the XGBoost model demonstrates excellent streamflow simulation capabilities, its black-box nature complicates the understanding of its internal mechanisms, which in turn affects its reliability (Dawid et al., 2022). Unlike the simple use of the XGBoost model for streamflow simulation and prediction, the SHAP method is utilized to analyze the interpretability of the XGBoost model output, thereby identifying the key controlling factors influencing streamflow variation. Through SHAP analysis, we find that the main controlling factors are highly consistent with the key variables outlined in classical hydrological theory (Wang et al., 2024; Yonaba et al., 2021). Among these variables, the most important feature is the ESB area, particularly when the proportion of the area exceeds 40%, indicating a positive contribution to streamflow. This positive contribution is because a larger area can accumulate more precipitation, which consequently increases streamflow (Huo et al., 2021; Zhang et al., 2017). Moreover, precipitation has emerged as an important factor influencing streamflow, whereas temperature has a relatively minor effect, which is consistent with the findings of numerous studies (Gou et al., 2022; Kormos et al., 2016; Woodhouse et al., 2016). The nonlinear effects of different land use types on streamflow show notable differences. The effects of grassland and cropland exhibit opposite trends. An increase in grassland area causes a reduction in streamflow (Li et al., 2017; Zhan et al., 2024), whereas an increase in cropland area exerts a positive regulatory effect on streamflow (Ahiablame et al., 2019; Zhang & Schilling, 2006). A moderate forestland proportion is beneficial for streamflow generation, as indicated in many studies, whereas excessive afforestation may lead to a decrease in streamflow (Ding et al., 2022; Xi et al., 2022). Importantly, the interaction effects between the different land use types suggest that various combinations of land use types exert complex regulatory effects on streamflow. A threshold effect is observed in the interactions between the different land use types. Specifically, when the proportion of one land use type reaches a certain threshold, the interaction effect between two land use types consistently contributes either positively or negatively to streamflow (Figure 11). This phenomenon may arise due to the dominant role of that land use type at this threshold. Like the interactions between various land use types, the interactions between precipitation and land use types notably vary, further emphasizing the dominant role of land use in shaping the hydrological responses of watersheds (Cavalcante et al., 2019; Xu et al., 2014).

4.3. Potential Applications in Watershed Water Resource Management

This study demonstrates the significant potential of combining hydrological models with machine learning methods, which can be applied in other watersheds worldwide to reveal the complex impacts of climate and land use changes on streamflow. Consistent with the findings of numerous studies, our quantitative assessment of the

relative contributions of climate and land use changes to streamflow reveals that the impact of climate change on watershed streamflow is relatively small, whereas land use changes have more significant effects (Cavalcante et al., 2019; Wang et al., 2019; Xu et al., 2014). This finding suggests that in watershed management, it is crucial to properly regulate land use types, particularly the proportions of cropland, grassland, and forestland, as well as their interactions. The nonlinear and interactive effects of land use types on streamflow highlight the need for balanced land use planning to maintain hydrological stability. Furthermore, the identified threshold effects provide practical guidance for defining land use limits, ensuring that no single type dominates to the detriment of the overall water resources. In the short term, climate change is difficult to control artificially, whereas land use adjustments can directly influence streamflow changes. In particular, within the context of water scarcity and ecological restoration tasks, the rational allocation of land types is crucial. In addition, this study reveals the importance rankings of precipitation, temperature, and various land use types within the watershed. Although barren land and water bodies occupy a relatively low proportion of the total area, their hydrological effects on streamflow should not be neglected. These land use types may exert significant regulatory effects, especially in small watersheds or specific areas. Therefore, in watershed water resource management and land use decision-making, addressing changes in major land use types and considering the roles of land use types with relatively low proportions are important. Overall, future watershed management should emphasize the multidimensional regulatory effects of land use, particularly under the dual pressures of climate and land use changes. Rationally planning land use patterns is essential for achieving sustainable management and optimizing watershed water resources.

4.4. Limitations and Recommendations

Although we have successfully achieved our research objectives, certain limitations should be acknowledged. This study focuses primarily on analyzing the impacts of significant driving factors, such as climate change and land use change, on streamflow variations. However, surface static attributes, such as the soil type and topographic slope, are not included in the analysis. Although the variations in these factors may be small, their intrinsic characteristics can influence streamflow generation, thereby affecting streamflow processes. Future research should involve further investigations into the influences of additional controlling factors on streamflow processes. In addition, we have examined only the controlling factors of streamflow variation at an annual scale. Future research should include analyses of these factors at high temporal resolutions, such as the monthly scale, to explore the impacts of various factors, including precipitation, temperature, the leaf area index, and other relevant variables, on streamflow. Finally, the framework developed in this study can be applied to future streamflow prediction. By utilizing land use prediction models, such as PLUS (Liang et al., 2021), FLUS (Liu et al., 2017), and CA-Markov (Sang et al., 2011), combined with different climate scenarios under CMIP6, researchers in the future can explore streamflow conditions under various climate and land use policy combinations, providing valuable scientific support for future water resource management.

5. Conclusions

In this study, we integrated a hydrological model with machine learning methods, utilizing the input and output data from the SWAT model to develop the XGBoost and RF models and compare their streamflow simulation performance characteristics. The SHAP method, along with scenario simulation, was applied to identify the factors controlling streamflow variation and to quantify their relative contributions. The main conclusions were as follows:

1. Both the XGBoost and RF models were effectively integrated with the SWAT model, and their streamflow simulation performance characteristics were compared, with XGBoost demonstrating superior performance. When trained with the input and output data of the SWAT model, comprising streamflow, climate, and land use data, the XGBoost model produced satisfactory streamflow simulation results. The mechanistically guided XGBoost model could become a promising tool for streamflow simulation.
2. The SHAP method could be used to effectively explain the output of the XGBoost model, thereby identifying the factors controlling streamflow variation. The area was the most important feature variable. With respect to climate change, precipitation had a greater impact on streamflow than did temperature, and when it exceeded 550 mm, it contributed positively to streamflow. Different land use types exerted nonlinear impacts on streamflow variation, with notable differences and threshold effects. Specifically, when the proportion of

- grassland was less than 50%, the proportion of cropland exceeded 20%, and the proportion of forestland varied between 30% and 50%, these land use types contributed positively to streamflow.
3. The interaction effects of the different land use types on streamflow exhibited nonlinear changes with notable differences. There were threshold effects in the interactions between different land use combinations, in which the interaction effect remained positive or negative once the area of one land use type reached a certain proportion. The interactions between precipitation and the various land use types notably differed, with precipitation playing a less dominant role in determining interaction effects.
 4. Quantitative analyses with different scenario settings revealed that the abrupt changes in watershed streamflow were driven primarily by significant land use changes, although climate change contributed as well. In terms of the ESBUs, the relative contribution of land use change ranged from 51.72% to 91.81%, whereas the contribution of climate change ranged from 8.19% to 48.28%. Land use changes across the entire watershed resulted in a streamflow decrease of 22.06 m³/s, whereas climate change led to a decrease of 17.54 m³/s, with relative contributions of 55.71% and 44.27%, respectively.

In this study, hydrological models were combined with machine learning methods, providing a more effective framework for streamflow simulation and analysis. This research not only offered precise support for water resource management in specific watersheds but also had broad application potential. This framework was found to be applicable to various global watersheds, and it helped address the water scarcity challenges caused by climate and land use changes.

Data Availability Statement

The digital elevation model (DEM) data were sourced from ASTER Global Digital Elevation Model Version 3 (<https://data.nasa.gov/dataset/aster-global-digital-elevation-model-v003-7ede1>) with a 30-m resolution. The soil data were derived from the Harmonized World Soil Database v2.0 (<https://www.fao.org/soils-portal/data-hub/en/>) with a scale of 1:1,000,000 and a resolution of 1 km. The land use data were obtained from the Annual Land Cover Product of China (<https://zenodo.org/records/8176941>), which used 30-m resolution data from 1985 to 2020. Meteorological data, including daily-scale information from 1980 to 2020, were obtained from the Climate Change Research Center (<https://ccrc.iap.ac.cn/resource/detail?id=228>). Additionally, the reconstruction of monthly streamflow data from 1980 to 2020 was obtained from the Haihe River Water Conservancy Commission, Ministry of Water Resources (<http://www.hwcc.gov.cn/>).

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References

- Ahiablame, L., Sheshukov, A. Y., Mosase, E., & Hong, J. (2019). Modelling the impacts of grassland to cropland conversion on river flow regimes in Skunk Creek watershed, Upper Midwest United States. *River Research and Applications*, 35(9), 1454–1465. <https://doi.org/10.1002/rra.3512>
- Anderson, B. J., Slater, L. J., Dadson, S. J., Blum, A. G., & Prosdocimi, I. (2022). Statistical attribution of the influence of urban and tree cover change on streamflow: A comparison of large sample statistical approaches. *Water Resources Research*, 58(5). <https://doi.org/10.1029/2021wr030742>
- Arnold, J. G., Moriasi, D. N., Gassman, P. W., Abbaspour, K. C., White, M. J., Srinivasan, R., et al. (2012). SWAT: Model use, calibration, and validation. *Transactions of the ASABE*, 55(4), 1491–1508. <https://doi.org/10.13031/2013.42256>
- Ben-Asher, J., & Berliner, P. (1994). *Runoff irrigation, Management of water use in agriculture* (pp. 126–154). Springer. https://doi.org/10.1007/978-3-642-78562-7_6
- Berghuijs, W. R., Larsen, J. R., Van Emmerik, T. H. M., & Woods, R. A. (2017). A global assessment of runoff sensitivity to changes in precipitation, potential evaporation, and other factors. *Water Resources Research*, 53(10), 8475–8486. <https://doi.org/10.1002/2017wr021593>
- Boucher, M. A., Quilty, J., & Adamowski, J. (2020). Data assimilation for streamflow forecasting using extreme learning machines and multilayer perceptrons. *Water Resources Research*, 56(6). <https://doi.org/10.1029/2019wr026226>
- Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32. <https://doi.org/10.1023/a:1010933404324>
- Cavalcante, R. B. L., Pontes, P. R. M., Souza, P. W. M., & De Souza, E. B. (2019). Opposite effects of climate and land use changes on the annual water balance in the Amazon arc of deforestation. *Water Resources Research*, 55(4), 3092–3106. <https://doi.org/10.1029/2019wr025083>
- Chen, T., & Guestrin, C. (2016). Xgboost: A scalable tree boosting system. *Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining*, 785–794. <https://doi.org/10.1145/2939672.2939785>
- Cheng, Z., & Yu, B. (2019). Effect of land clearing and climate variability on streamflow for two large basins in Central Queensland, Australia. *Journal of Hydrology*, 578, 124041. <https://doi.org/10.1016/j.jhydrol.2019.124041>
- Das, A., & Rad, P. (2020). Opportunities and challenges in explainable artificial intelligence (xai): A survey. arXiv preprint arXiv:2006.11371. <https://doi.org/10.48550/arXiv.2006.11371>
- Dawid, A., Huembeli, P., Tomza, M., Lewenstein, M., & Dauphin, A. (2022). Hessian-based toolbox for reliable and interpretable machine learning in physics. *Machine Learning-Science And Technology*, 3(1), 015002. <https://doi.org/10.1088/2632-2153/ac338d>
- Dey, P., & Mishra, A. (2017). Separating the impacts of climate change and human activities on streamflow: A review of methodologies and critical assumptions. *Journal of Hydrology*, 548, 278–290. <https://doi.org/10.1016/j.jhydrol.2017.03.014>

- Ding, B., Zhang, Y., Yu, X., Jia, G., Wang, Y., Wang, Y., et al. (2022). Effects of forest cover type and ratio changes on runoff and its components. *International Soil and Water Conservation Research*, 10(3), 445–456. <https://doi.org/10.1016/j.iswcr.2022.01.006>
- Ding, B. B., Zhang, J., Zheng, P., Li, Z., Wang, Y., Jia, G., & Yu, X. (2024). Water security assessment for effective water resource management based on multi-temporal blue and green water footprints. *Journal of Hydrology*, 632, 130761. <https://doi.org/10.1016/j.jhydrol.2024.130761>
- Gan, G., Liu, Y., & Sun, G. (2021). Understanding interactions among climate, water, and vegetation with the Budyko framework. *Earth-Science Reviews*, 212, 103451. <https://doi.org/10.1016/j.earscirev.2020.103451>
- Gao, H., Li, Q., Xiong, G., Li, B., Zhang, J., & Meng, Q. (2023). Quantitative assessment of hydrological response to vegetation change in the upper reaches of Luanhe River with the modified Budyko framework. *Frontiers in Ecology and Evolution*, 11, 1178231. <https://doi.org/10.3389/fevo.2023.1178231>
- Gassman, P. W., Sadeghi, A. M., & Srinivasan, R. (2014). Applications of the SWAT model special section: Overview and insights. *Journal of Environmental Quality*, 43(1), 1–8. <https://doi.org/10.2134/jeq2013.11.0466>
- Gou, J. J., Miao, C., Duan, Q., Zhang, Q., Guo, X., & Su, T. (2022). Seasonality and impact factor analysis of streamflow sensitivity to climate change across China. *Earth's Future*, 10(11). <https://doi.org/10.1029/2022ef003062>
- Huo, J. Y., Liu, C. J., Yu, X. X., Jia, G. D., & Chen, L. H. (2021). Effects of watershed char and climate variables on annual runoff in different climatic zones in China. *Science Of the Total Environment*, 754, 142157. <https://doi.org/10.1016/j.scitotenv.2020.142157>
- Jiang, C., Xiong, L., Wang, D., Liu, P., Guo, S., & Xu, C.-Y. (2015). Separating the impacts of climate change and human activities on runoff using the Budyko-type equations with time-varying parameters. *Journal of Hydrology*, 522, 326–338. <https://doi.org/10.1016/j.jhydrol.2014.12.060>
- Kormos, P. R., Luce, C. H., Wenger, S. J., & Berghuijs, W. R. (2016). Trends and sensitivities of low streamflow extremes to discharge timing and magnitude in Pacific Northwest mountain streams. *Water Resources Research*, 52(7), 4990–5007. <https://doi.org/10.1002/2015wr018125>
- Kratzert, F., Klotz, D., Herrnegger, M., Sampson, A. K., Hochreiter, S., & Nearing, G. S. (2019). Toward improved predictions in ungauged basins: Exploiting the power of machine learning. *Water Resources Research*, 55(12), 11344–11354. <https://doi.org/10.1029/2019wr026065>
- Lane, S. N. (2017). Natural flood management. *Wiley Interdisciplinary Reviews: Water*, 4(3), e1211. <https://doi.org/10.1002/wat2.1211>
- Li, J., Liu, D., Wang, T., Li, Y., Wang, S., Yang, Y., et al. (2017). Grassland restoration reduces water yield in the headstream region of Yangtze River. *Scientific Reports*, 7(1), 2162. <https://doi.org/10.1038/s41598-017-02413-9>
- Li, M., Zhang, Y. Q., Wallace, J., & Campbell, E. (2020). Estimating annual runoff in response to forest change: A statistical method based on random forest. *Journal of Hydrology*, 589, 125168. <https://doi.org/10.1016/j.jhydrol.2020.125168>
- Li, Y., Piao, S., Li, L. Z. X., Chen, A., Wang, X., Ciais, P., et al. (2018). Divergent hydrological response to large-scale afforestation and vegetation greening in China. *Science Advances*, 4(5). <https://doi.org/10.1126/sciadv.aar4182>
- Liang, X., Guan, Q., Clarke, K. C., Liu, S., Wang, B., & Yao, Y. (2021). Understanding the drivers of sustainable land expansion using a patch-generating land use simulation (PLUS) model: A case study in Wuhan, China. *Computers, Environment and Urban Systems*, 85, 101569. <https://doi.org/10.1016/j.compenvurbsys.2020.101569>
- Liang, Z. Y., Zou, R., Chen, X., Ren, T., Su, H., & Liu, Y. (2020). Simulate the forecast capacity of a complicated water quality model using the long short-term memory approach. *Journal of Hydrology*, 581, 124432. <https://doi.org/10.1016/j.jhydrol.2019.124432>
- Liu, X., Liang, X., Li, X., Xu, X., Ou, J., Chen, Y., et al. (2017). A future land use simulation model (FLUS) for simulating multiple land use scenarios by coupling human and natural effects. *Landscape and Urban Planning*, 168, 94–116. <https://doi.org/10.1016/j.landurbplan.2017.09.019>
- Lundberg, S. (2017). A unified approach to interpreting model predictions. *arXiv preprint arXiv:1705.07874*.
- Luo, S., Yang, Q., Wu, H., Liu, J., Wang, G., Wang, Y., & Yang, Y. (2019). Dynamic attribution analysis for runoff change integrating Landsat-derived land use dynamics with SWAT model. *IGARSS 2019-2019 IEEE International Geoscience and Remote Sensing Symposium* (pp. 6413–6416). IEEE. <https://doi.org/10.1109/igarss.2019.8900316>
- Lye, D. J. (2009). Rooftop runoff as a source of contamination: A review. *Science of the total environment*, 407(21), 5429–5434. <https://doi.org/10.1016/j.scitotenv.2009.07.011>
- Nearing, G. S., Kratzert, F., Sampson, A. K., Pelissier, C. S., Klotz, D., Frame, J. M., et al. (2021). What role does hydrological science play in the age of machine learning? *Water Resources Research*, 57(3). <https://doi.org/10.1029/2020wr028091>
- Oki, T., & Kanae, S. (2006). Global hydrological cycles and world water resources. *Science*, 313(5790), 1068–1072. <https://doi.org/10.1126/science.1128845>
- Rajib, M. A., Merwade, V., Kim, I. L., Zhao, L., Song, C., & Zhe, S. (2016). SWATShare—A web platform for collaborative research and education through online sharing, simulation and visualization of SWAT models. *Environmental Modelling & Software*, 75, 498–512. <https://doi.org/10.1016/j.envsoft.2015.10.032>
- Ran, Q., Su, D., Li, P., & He, Z. (2012). Experimental study of the impact of rainfall characteristics on runoff generation and soil erosion. *Journal of Hydrology*, 424, 99–111. <https://doi.org/10.1016/j.jhydrol.2011.12.035>
- Sang, L., Zhang, C., Yang, J., Zhu, D., & Yun, W. (2011). Simulation of land use spatial pattern of towns and villages based on CA-Markov model. *Mathematical and Computer Modelling*, 54(3–4), 938–943. <https://doi.org/10.1016/j.mcm.2010.11.019>
- Sharma, A., Patel, P., & Sharma, P. J. (2022). Influence of climate and land-use changes on the sensitivity of SWAT model parameters and water availability in a semi-arid river basin. *Catena*, 215, 106298. <https://doi.org/10.1016/j.catena.2022.106298>
- Tang, S. L., Sun, F., Liu, W., Wang, H., Feng, Y., & Li, Z. (2023). Optimal postprocessing strategies with LSTM for global streamflow prediction in ungauged basins. *Water Resources Research*, 59(7). <https://doi.org/10.1029/2022wr034352>
- Wang, D. D., Yu, X. X., Jia, G. D., & Wang, H. N. (2019). Sensitivity analysis of runoff to climate variability and land-use changes in the Haihe Basin mountainous area of north China. *Agriculture, Ecosystems & Environment*, 269, 193–203. <https://doi.org/10.1016/j.agee.2018.09.025>
- Wang, S., Liu, Y., Wang, W., Zhao, G., & Liang, H. (2024). Interpretable machine learning guided by physical mechanisms reveals drivers of runoff under dynamic land use changes. *Journal of Environmental Management*, 367, 121978. <https://doi.org/10.1016/j.jenvman.2024.121978>
- Wang, S., Peng, H., Hu, Q., & Jiang, M. (2022). Analysis of runoff generation driving factors based on hydrological model and interpretable machine learning method. *Journal of Hydrology-Regional Studies*, 42, 101139. <https://doi.org/10.1016/j.ejrh.2022.101139>
- Woodhouse, C. A., Pederson, G. T., Morino, K., McAfee, S. A., & McCabe, G. J. (2016). Increasing influence of air temperature on upper Colorado River streamflow. *Geophysical Research Letters*, 43(5), 2174–2181. <https://doi.org/10.1002/2015gl067613>
- Wu, L., Zhang, X., Hao, F., Wu, Y., Li, C., & Xu, Y. (2020). Evaluating the contributions of climate change and human activities to runoff in typical semi-arid area, China. *Journal of Hydrology*, 590, 125555. <https://doi.org/10.1016/j.jhydrol.2020.125555>
- Xi, Y., Peng, S., Liu, G., Ducharme, A., Ciais, P., Prigent, C., et al. (2022). Trade-off between tree planting and wetland conservation in China. *Nature Communications*, 13(1), 1967. <https://doi.org/10.1038/s41467-022-29616-7>
- Xu, H., Ren, Y., Zheng, H., Ouyang, Z., & Jiang, B. (2020). Analysis of runoff trends and drivers in the Haihe River Basin, China. *International Journal of Environmental Research and Public Health*, 17(5), 1577. <https://doi.org/10.3390/ijerph17051577>

- Xu, X., Yang, D., Yang, H., & Lei, H. (2014). Attribution analysis based on the Budyko hypothesis for detecting the dominant cause of runoff decline in Haihe basin. *Journal of Hydrology*, 510, 530–540. <https://doi.org/10.1016/j.jhydrol.2013.12.052>
- Yang, H. B., Yang, D. W., & Hu, Q. F. (2014). An error analysis of the Budyko hypothesis for assessing the contribution of climate change to runoff. *Water Resources Research*, 50(12), 9620–9629. <https://doi.org/10.1002/2014wr015451>
- Yonaba, R., Biau, A. C., Koita, M., Tazen, F., Mounirou, L. A., Zouré, C. O., et al. (2021). A dynamic land use/land cover input helps in picturing the Sahelian paradox: Assessing variability and attribution of changes in surface runoff in a Sahelian watershed. *Science Of the Total Environment*, 757, 143792. <https://doi.org/10.1016/j.scitotenv.2020.143792>
- Zeng, R., & Cai, X. (2016). Climatic and terrestrial storage control on evapotranspiration temporal variability: Analysis of river basins around the world. *Geophysical Research Letters*, 43(1), 185–195. <https://doi.org/10.1002/2015gl066470>
- Zhan, H., Yu, D., Wang, L., Zhang, J., Xu, M., Fang, X., et al. (2024). Stronger influences of grassland growth than grassland area on hydrological processes in the source region of the Yellow River. *Journal of Hydrology*, 642, 131886. <https://doi.org/10.1016/j.jhydrol.2024.131886>
- Zhang, C., & Ma, Y. (2012). *Ensemble machine learning* (Vol. 144). Springer. <https://doi.org/10.1007/978-1-4419-9326-7>
- Zhang, M. F., Liu, N., Harper, R., Li, Q., Liu, K., Wei, X., et al. (2017). A global review on hydrological responses to forest change across multiple spatial scales: Importance of scale, climate, forest type and hydrological regime. *Journal of Hydrology*, 546, 44–59. <https://doi.org/10.1016/j.jhydrol.2016.12.040>
- Zhang, W., Villarini, G., Vecchi, G. A., & Smith, J. A. (2018). Urbanization exacerbated the rainfall and flooding caused by hurricane Harvey in Houston. *Nature*, 563(7731), 384–388. <https://doi.org/10.1038/s41586-018-0676-z>
- Zhang, Y.-K., & Schilling, K. (2006). Increasing streamflow and baseflow in Mississippi River since the 1940 s: Effect of land use change. *Journal of Hydrology*, 324(1–4), 412–422. <https://doi.org/10.1016/j.jhydrol.2005.09.033>
- Zhao, F., Wu, Y., Qiu, L., Sun, Y., Sun, L., Li, Q., et al. (2018). Parameter uncertainty analysis of the SWAT model in a mountain-loess transitional watershed on the Chinese Loess Plateau. *Water*, 10(6), 690. <https://doi.org/10.3390/w10060690>
- Zounemat-Kermani, M., Batelaan, O., Fadaee, M., & Hinkelmann, R. (2021). Ensemble machine learning paradigms in hydrology: A review. *Journal of Hydrology*, 598, 126266. <https://doi.org/10.1016/j.jhydrol.2021.126266>

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